🤖 CRYPTO INTELLIGENCE AGENT — MASTER PROMPT COLLECTION

> Complete build guide: all prompts, all levels, all systems in one file. Use these prompts with any LLM-powered coding tool (Replit, Cursor, Claude Code, etc.)

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PART 1: FOUNDATION

Levels 1–10: Building the Core Bot

LEVEL 1 — Foundation Bot (Weekend Build)

```
CONTEXT:
I want to build a personal crypto assistant Telegram bot from scratch.
I have zero coding experience. I want to use Python.
I have a Telegram account and can get a bot token from BotFather.
I have an Anthropic API key for Claude.
```

```
TASK:
Build a complete working Telegram bot that:
1. Responds to any message using Claude AI
2. Knows my crypto portfolio (I'll configure it in a JSON file)
3. Can answer questions like "what is my ETH position worth?"
4. Has a /start command that introduces itself
5. Has a /help command listing what it can do
6. Stores conversation history so Claude remembers context

REQUIREMENTS:
- Use python-telegram-bot library
- Use anthropic SDK for Claude
- Store portfolio in config.json file
- Store conversation history in SQLite database
- Keep the system prompt focused on crypto assistance
- Bot should feel like a knowledgeable crypto advisor

PORTFOLIO CONFIG FORMAT (config.json):
{
  "portfolio": [
    {"symbol": "BTC", "quantity": 0.5, "avg_buy_price": 45000},
    {"symbol": "ETH", "quantity": 2.0, "avg_buy_price": 2800},
    {"symbol": "SOL", "quantity": 10, "avg_buy_price": 95}
  ],
  "risk_tolerance": "medium",
  "base_currency": "USD"
}

CONSTRAINTS:
- Single file to start (bot.py)
- No paid APIs except Anthropic (already have key)
- Must work on Replit free tier
- Keep it simple — no complex architecture yet

SUCCESS LOOKS LIKE:
- I send "what's my portfolio worth?" and get a response
  with my positions and current values (even if prices are approximate)
- I send "explain DeFi to me" and get a helpful explanation
- Conversation context is remembered within a session
```

LEVEL 2 — Real-Time Prices

```
CONTEXT:
```

```
I have a working Telegram crypto bot from Level 1.
It uses Python, python-telegram-bot, Anthropic Claude.
Now I want to add real price data.

TASK:
Add real-time price fetching using the Binance public API.
No API key needed for basic price data.

BINANCE PRICE ENDPOINT:
https://api.binance.com/api/v3/ticker/price?symbol=BTCUSDT

REQUIREMENTS:
1. Create market.py module with:
   - get_price(symbol) function
   - get_multiple_prices(symbols) function
   - Cache prices for 60 seconds (don't hit API on every message)

2. Add /price command:
   /price BTC → shows current BTC price with 24h change
   /price BTC ETH SOL → shows multiple prices

3. Update portfolio display to show REAL current prices:
   When user asks about portfolio, fetch live prices
   Calculate real P&L for each position

4. Inject current prices into Claude's context:
   Before every Claude response, add:
   "Current prices: BTC=$X, ETH=$X, SOL=$X"
   So Claude can reference real data in answers

CONSTRAINTS:
- Handle API failures gracefully (show cached price or "unavailable")
- Don't make more than 1 request per symbol per minute
- Keep market.py as a separate file from bot.py

SUCCESS LOOKS LIKE:
- /price BTC returns current real price instantly
- Asking "what is my portfolio worth?" returns accurate current value
- Claude can answer "is BTC up or down today?" with real data
```

LEVEL 3 — Price Alerts

```
CONTEXT:
My crypto bot now has real-time prices via Binance API.
```

```
I want to add price alerts that notify me automatically.

TASK:
Build a complete price alert system.

COMMAND FORMAT:
/alert BTC above 100000   → alert when BTC goes above $100,000
/alert ETH below 3000     → alert when ETH drops below $3,000
/alerts                    → show all my active alerts
/cancelalert 3            → cancel alert #3

REQUIREMENTS:
1. Create alerts.py with:
   - Alert storage in SQLite (survives bot restarts)
   - Background checker that runs every 60 seconds
   - Checks all active alerts against current prices
   - Sends Telegram message when alert triggers
   - Marks alert as triggered (doesn't fire repeatedly)

2. Alert message format when triggered:
   🚨 ALERT TRIGGERED
   BTC crossed above $100,000!
   Current price: $100,247
   Alert set: 3 days ago
   Your position: 0.5 BTC ($50,123)

3. /alerts command shows:
   Active Alerts:
   #1 BTC > $100,000 (currently $97,500 — 2.6% away)
   #2 ETH < $3,000 (currently $3,100 — 3.2% away)

CONSTRAINTS:
- Store alerts in database so they survive restart
- Support at least 20 simultaneous alerts
- Don't alert more than once per trigger event
- Background checker must not block the main bot

SUCCESS:
- Set alert, go to sleep, wake up to notification
- /alerts shows distance to each alert target
```

LEVEL 4 — Market Intelligence

```
CONTEXT:
```

```
Bot has prices and alerts. Now give Claude real market context.

TASK:
Build a market data module that gives Claude a comprehensive
market overview to work with.

DATA TO FETCH (all free APIs):

1. CoinGecko Global Market:
   https://api.coingecko.com/api/v3/global
   Returns: total market cap, BTC dominance, 24h volume

2. Fear & Greed Index:
   https://api.alternative.me/fng/?limit=1
   Returns: score (0-100) and classification

3. Top 10 coins with 24h change:
   https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd&order=market_ca

TASK:
Create market.py additions:
- get_market_overview() combining all above
- Cache market data for 10 minutes
- Format it for Claude context injection

ADD COMMAND /market:
Shows formatted overview:

📊 MARKET OVERVIEW
Total Market Cap: $2.1T (+1.8% 24h)
BTC Dominance: 54.2%
Fear & Greed: 67 (Greed)

TOP MOVERS:
📈 BTC: $97,500 (+2.1%)
📈 ETH: $3,100 (+1.4%)
📉 SOL: $182 (-0.8%)
[...top 10]

INJECT INTO EVERY CLAUDE CALL:
Add market overview to system context so Claude
always has current market data when answering questions.
```

LEVEL 5 — Trading Journal

```
CONTEXT:
Bot has market data and alerts. Now add a trading journal
so the bot remembers my trading decisions and learns from them.

TASK:
Build a complete trading journal system.

COMMANDS:
/journal Bought 0.5 ETH at 3100, RSI bounce setup
/journal Sold BTC position, took 15% profit, felt nervous
/journal note Market looking weak, reducing exposure
/myjournal          → shows last 10 entries
/myjournal ETH      → shows entries mentioning ETH
/myjournal search "RSI bounce"  → searches entries

JOURNAL ENTRY AUTO-EXTRACTION:
When I log a trade, Claude automatically extracts:
- Action (buy/sell/hold/note)
- Coin mentioned
- Price if mentioned
- Sentiment (bullish/bearish/neutral — from tone)
- Key reasoning (1 sentence summary)

DATABASE TABLE: journal
- id, timestamp, raw_entry (my words)
- extracted_action, extracted_coin
- extracted_price, sentiment, reasoning_summary
- outcome (filled in later by user)

CLAUDE READS MY JOURNAL:
Add last 5 journal entries to Claude's context.
So when I ask "what have I been thinking about ETH?"
Claude can reference my actual journal.

FORMAT FOR /myjournal:
📔 YOUR TRADING JOURNAL

Feb 23 3:47 PM:
"Bought 0.5 ETH at 3100, RSI bounce setup"
Action: Buy | Coin: ETH | Sentiment: Bullish

Feb 21 11:22 AM:
"Sold BTC, took 15% profit"
Action: Sell | Coin: BTC | Sentiment: Cautious
```

LEVEL 6 — Daily Briefing

```
CONTEXT:
Bot has journal, market data, portfolio tracking.
Now automate a daily morning briefing.

TASK:
Every morning at 8:00 AM (my timezone), automatically send
a comprehensive market briefing to Telegram.

BRIEFING CONTENT:
1. Overnight price movements for my portfolio coins
2. Overall market sentiment (Fear & Greed)
3. Any alerts that are close to triggering (<5% away)
4. Claude's 3-sentence market commentary
5. My top priority for the day (based on portfolio state)

FORMAT:
☀️ GOOD MORNING — CRYPTO BRIEFING
Wednesday, February 23

YOUR PORTFOLIO:
BTC: $97,500 (+2.3% overnight) ↑
ETH: $3,100 (-0.8% overnight) ↓
SOL: $182 (+1.1% overnight) ↑
Total: $58,350 (+1.4% overnight)

MARKET MOOD: 67 — Greed 😏

ALERT WATCH:
⚠️ ETH approaching alert at $3,000 (3.2% away)

AI COMMENTARY:
[Claude's 3-sentence assessment of current conditions]

TODAY'S FOCUS:
[Claude's one recommended area of attention]

SETUP:
- Use APScheduler for scheduling
- Detect timezone from first interaction or ask user to set it
- Command /briefingtime 9am → changes delivery time
- Command /briefing → get briefing on demand anytime
```

LEVEL 7 — Advanced Alerts

```
CONTEXT:
I have basic price alerts. I want smarter, more complex alerts.

TASK:
Upgrade alerts.py with advanced condition types.

NEW ALERT TYPES:

1. PERCENTAGE CHANGE ALERT:
   /alert BTC change 5%
   → Alert if BTC moves 5% in either direction within 1 hour

2. PORTFOLIO ALERTS:
   /alert portfolio down 10%
   → Alert if total portfolio drops 10% from current value

3. MULTI-CONDITION ALERT (AND):
   /alert BTC above 95000 AND ETH above 3000
   → Only alert if BOTH conditions are true simultaneously

4. RSI ALERT:
   /alert ETH rsi below 30
   → Alert when ETH RSI drops below 30 on 4h chart
   (Fetch RSI from Binance kline data, calculate yourself)

5. VOLUME ALERT:
   /alert BTC volume spike 3x
   → Alert when BTC volume is 3x above its 7-day average

COMMAND FORMAT:
/smartalert [description in plain English]
→ Claude interprets and creates the right alert type

Example:
/smartalert tell me if BTC drops 5% in an hour or breaks above 100k
→ Creates two alerts from this one command

ALERT MANAGEMENT:
/alerts → shows all with type icons
🎯 Price | 📊 Portfolio | 🔢 RSI | 📦 Volume | 🔗 Multi

/alerthistory → shows last 20 triggered alerts and outcomes
```

LEVEL 8 — Wallet Watcher

```
CONTEXT:
Adding on-chain monitoring for wallets I want to track.

TASK:
Build a wallet monitoring system using free blockchain APIs.

SETUP COMMAND:
/watchwallet 0x742d35cc6634c0532925a3b844bc454e4438f44e "My ETH wallet"
/watchwallet bc1q... "My BTC wallet"
/mywatched → shows all watched wallets

MONITORING (every 5 minutes):
For ETH wallets (Etherscan free API):
- Check for new transactions
- If new transaction found, alert:

  👛 WALLET ACTIVITY DETECTED
  Wallet: "My ETH wallet" (0x742d...f44e)

  New transaction:
  Type: Received ETH
  Amount: 2.5 ETH ($7,750)
  From: 0x1234...5678
  Time: 2 minutes ago
  TX: [link to etherscan]

BALANCE TRACKING:
/balance → shows current balance of all watched wallets
Store historical balances, show trend.

PORTFOLIO INTEGRATION:
If I add my own wallet, option to auto-sync
positions from on-chain data into portfolio.

FREE API:
Etherscan: https://api.etherscan.io/api
Get free key at etherscan.io (100k calls/day)
```

LEVEL 9 — AI Strategy Advisor

```
CONTEXT:
Now I want Claude to proactively analyze my situation
```

and give strategic advice, not just answer questions.

TASK:
Build an AI strategy advisor that runs deep analysis on demand.

COMMAND: /analyze
Triggers comprehensive analysis:

Step 1: Gather all data:
- Current portfolio with real prices and P&L
- Last 30 days of my journal entries
- Current market conditions
- Technical indicators for my holdings
- Recent news sentiment

Step 2: Send everything to Claude with prompt:
"You are analyzing this trader's complete situation.
Portfolio: [data]
Recent journal: [entries]
Market conditions: [data]
Technical signals: [indicators]

Provide:
1. Portfolio health assessment (what's working, what isn't)
2. Risk assessment (what could hurt them)
3. One specific opportunity you see
4. One specific risk you're concerned about
5. Recommended focus for next 7 days

Be specific. Reference their actual numbers.
Keep total response under 300 words."

Step 3: Send formatted analysis to Telegram.

ADDITIONAL COMMANDS:
/advisor ETH → focused analysis on just ETH
/advisor risk → focused portfolio risk assessment
/advisor weekly → weekly strategy review (also scheduled Sunday)

SCHEDULING:
Sunday 6pm: automatic weekly review
(ask Claude to review week's trades from journal
and give performance feedback)

LEVEL 10 — Security & Polish

```
CONTEXT:
Bot is working. Now make it secure, reliable, and production-ready.

TASK:
Security hardening, error handling, and reliability improvements.

SECURITY:
1. User authorization:
   Only YOUR Telegram user ID can use the bot.
   Add to .env: ALLOWED_USER_ID=123456789
   Every command: check if sender is authorized.
   Unauthorized users get: "This is a private bot."

2. Rate limiting:
   Max 20 messages per minute from any user.
   Prevents runaway API costs if something goes wrong.

3. Sensitive data auto-delete:
   Messages containing portfolio values:
   Auto-delete after 5 minutes (Telegram supports this).
   Add "(auto-deleting in 5 min)" notice.

4. API key validation on startup:
   Test all API keys on startup.
   If any key fails, log warning but continue.
   Alert via Telegram if Anthropic key fails.

RELIABILITY:
5. Global error handler:
   Every command wrapped in try/except.
   Errors logged to file AND sent to Telegram as admin alert.
   User gets friendly "something went wrong" message.

6. Health check command:
   /health → tests all systems and reports:
   ✅ Database: OK
   ✅ Anthropic API: OK
   ✅ Binance API: OK
   ✅ Alerts running: OK (12 active)
   ✅ Scheduled tasks: OK

7. Graceful shutdown:
   On SIGTERM: finish current request, save state, exit cleanly.
```

```
8. Auto-restart on crash:
   Add process manager (use Replit's always-on, or systemd)

POLISH:
9. /stats command:
   Messages sent today: 47
   API calls made: 123
   Uptime: 14 days, 6 hours
   Alerts triggered this week: 8

10. Logging:
    Every action logged with timestamp.
    Log rotation: keep 7 days of logs.
```

PART 2: INTELLIGENCE LAYER

Levels 11–20

LEVEL 11 — Market Scanner

```
Create autonomous/scanner.py

This is a background service that continuously
scans the entire crypto market for significant
events and opportunities WITHOUT me asking.
It should run every 5 minutes.

SCAN 1: BIG MOVERS DETECTOR
Every 5 minutes:
- Fetch top 100 coins from CoinGecko
- Find any coin that moved more than 8% in the last hour
- Find any coin in my watchlist that moved more than 5%
- If found, analyze WHY using Claude (2 sentences)
- Send Telegram notification:

  🚨 SIGNIFICANT MOVE DETECTED
  📈 PEPE +12.3% in last hour
  Current: $0.0000187 | Volume: 3.2x above normal
  🤖 AI Take: [Claude's 2-sentence analysis]
  /analyze PEPE for full analysis
```

```
SCAN 2: VOLUME ANOMALY DETECTOR
- Compare current 1h volume to 24h average hourly volume
- Alert if any top-100 coin has 5x+ normal volume

SCAN 3: MARKET STRUCTURE MONITOR
Every 15 minutes:
- Fear & Greed changed 10+ points in 2 hours?
- BTC dominance shifted 1%+ in 4 hours?
- Total market cap dropped 3%+ in 1 hour?

SCAN 4: CORRELATION BREAK DETECTOR
Every hour:
- Check if ETH is NOT following BTC
- Alert: "BTC/ETH correlation breaking down"

SCAN 5: PORTFOLIO HEALTH CHECK
Every 30 minutes:
- Any position down more than 7% from yesterday?
- Total portfolio down more than 5%?

SMART FILTERING:
- Don't send same alert twice within 4 hours
- Between midnight-7am: only CRITICAL alerts
- /scanner command shows status and recent findings
- /scanner off/on to pause/resume
```

PROMPT 11.2 — Opportunity Finder

```
Add opportunity detection to the scanner.

OPPORTUNITY SCAN 1: OVERSOLD QUALITY COINS
Every hour:
- Fetch RSI for top 50 coins (Binance kline data)
- Find coins with RSI below 35 on 4h chart
- Filter: top 30 market cap only
- Check: above 200-day SMA AND volume declining
- If both true, send opportunity alert:

  💡 OPPORTUNITY RADAR
  ETH showing oversold on 4h
  RSI: 32.4 | Above 200 SMA: Yes ✅ | Volume: Declining ✅
  /ta ETH 4h for full analysis

OPPORTUNITY SCAN 2: FUNDING RATE EXTREMES
```

```
Every hour via Binance:
https://fapi.binance.com/fapi/v1/fundingRate?symbol={symbol}USDT&limit=1

Funding > 0.08% per 8h = overheated longs
Funding < -0.04% per 8h = short squeeze conditions

OPPORTUNITY SCAN 3: NEW 90-DAY HIGHS
Track 90-day high for top 20 coins.
Alert when any breaks to new 90-day high.

OPPORTUNITY SCAN 4: SECTOR ROTATION
Organize coins into sectors (L1s, DeFi, AI, Gaming, L2s).
Every 4 hours: calculate average performance per sector.
If one sector 2x outperforming: rotation alert.

/opportunities command shows recent opportunity alerts.
```

LEVEL 12 — On-Chain Intelligence

```
Create data/onchain.py

FREE ON-CHAIN DATA SOURCES:

1. ETHERSCAN (free key from etherscan.io):
   get_eth_gas_prices() → slow/standard/fast + USD costs
   get_large_eth_transactions(min_eth=100) → whale moves
   get_wallet_balance(address) → ETH + top tokens

2. BITCOIN (blockchain.info free):
   get_btc_mempool_info() → congestion level + fees
   get_btc_network_stats() → hash rate, difficulty, TPS

3. DEFILLAMA (free, no key):
   get_protocol_tvl(protocol_name)
   get_tvl_changes() → top 20 protocols + 24h change
   get_stablecoin_data() → USDT/USDC supply + minting

COMBINED FUNCTION: build_onchain_summary()
Returns formatted summary:

"ON-CHAIN INTELLIGENCE:
ETHEREUM: Gas 12 gwei (Low) ✅
BITCOIN: Mempool congested (45k unconfirmed txns)
DEFI: TVL $98.7B (+2.1% 24h), Aave biggest inflow
```

```
STABLECOINS: USDC supply +$200M minted today"

INJECT INTO CLAUDE: Add on-chain summary to every call context.

COMMANDS:
/onchain → full on-chain summary
/gas → quick gas price check with USD costs per operation
```

LEVEL 13 — News & Sentiment Engine

```
Create data/news.py

FREE NEWS SOURCES:
1. CryptoPanic API: https://cryptopanic.com/api/v1/posts/?auth_token=FREE&public=
2. CoinGecko News: https://api.coingecko.com/api/v3/news
3. RSS Feeds (feedparser): CoinDesk, CoinTelegraph, Decrypt, The Block

FUNCTIONS:
fetch_latest_news(limit=20) → deduplicated, sorted by recency
get_news_for_symbol(symbol) → filtered by coin mention

analyze_news_sentiment(articles):
Send titles to Claude:
"Rate sentiment as VERY_BEARISH/BEARISH/NEUTRAL/BULLISH/VERY_BULLISH.
Identify: top story, any regulatory news, any critical news.
Return JSON: {sentiment, score, top_story, regulatory, critical}"

generate_news_briefing() → formatted sentiment dashboard:
📰 CRYPTO NEWS SENTIMENT
Overall: 🟢 Bullish (7.2/10)
🔥 Top Story: [headline]
⚠️ Regulatory: [if any]

AUTOMATED: Every 30 min check news about my portfolio coins.
Alert if negative news found:
📰 NEWS ALERT — YOUR PORTFOLIO
ETH in 3 new articles | Sentiment: Bearish ⚠️

COMMANDS:
/news → current sentiment briefing
/news BTC → Bitcoin-specific news
/sentiment → quick market sentiment dashboard
```

LEVEL 14 — Research Agent + Portfolio Optimizer

```
Create intelligence/research_agent.py

COMMAND: /research BTC
Runs autonomous multi-step research:

Step 1: Announce "Starting research... taking 2-3 minutes"
Step 2: Gather simultaneously:
  - Price history (24h/7d/30d changes, ATH distance)
  - Market data (market cap, supply, FDV)
  - News (last 7 days + sentiment)
  - On-chain (token holders, whale movements)
  - DeFi data (TVL, revenue from DeFiLlama)

Step 3: Claude synthesis with prompt:
"Write a research report covering:
1. Executive Summary (3 sentences)
2. Technical Picture
3. Fundamental Strength
4. Catalysts & Risks
5. Smart Money Signals
6. Verdict: STRONG BUY / BUY / NEUTRAL / AVOID / STRONG AVOID
Be specific with numbers. Under 400 words."

Step 4: Send report. Save to journal.

COMMANDS:
/research BTC → full research
/compare BTC ETH SOL → research all, rank by opportunity
/watchlist → manage research watchlist (auto-research weekly)

---

Create intelligence/portfolio_optimizer.py

/optimize → runs full portfolio analysis:
- Concentration risk (any coin >40%?)
- Correlation analysis (BTC/ETH both heavy = not diversified)
- Sector exposure (what are you missing?)
- Dead weight detection (down 30%+ with declining volume)
- Opportunity cost (vs just holding BTC)
- Claude synthesis: top 3 strengths, top 3 risks, one action

/risk → quick risk dashboard with gauges
/rebalance BTC:40 ETH:30 SOL:20 CASH:10 → exact trade plan
```

LEVEL 15 — Professional Technical Analysis

```
Upgrade data/technical.py with 9 professional indicators.
All data from Binance kline API (free):
https://api.binance.com/api/v3/klines?symbol={symbol}USDT&interval={interval}&lim

ADD THESE INDICATORS:

1. MACD: 12/26 EMA, Signal 9, Histogram
   Output: crossover status, momentum direction

2. BOLLINGER BANDS: 20 SMA ± 2 SD
   Output: upper/lower/middle, squeeze detection

3. STOCHASTIC RSI: RSI of RSI (0-100)
   Below 20 = oversold, above 80 = overbought

4. VWAP: Volume Weighted Average Price
   Above = bullish, below = bearish

5. ATR (14): Average True Range for stop sizing
   Output: suggested 2x ATR stop loss distance

6. FIBONACCI: Find swing high/low, calculate 23.6/38.2/50/61.8/78.6%
   Output: nearest fib level to current price

7. ADX: Trend strength 0-100
   <25 ranging, 25-50 moderate, 50+ strong trend

8. MULTI-TIMEFRAME ALIGNMENT:
   Check RSI on 1h, 4h, 1d simultaneously
   "3/3 Bullish = Strong Conviction"

FULL ANALYSIS COMMAND: /ta BTC
Outputs complete formatted analysis with all 8 indicators,
key levels, volume analysis, and Claude's 2-sentence read.

/ta BTC 1h → specific timeframe analysis
```

LEVEL 16 — Performance Tracking & Adaptive Memory

```
Create intelligence/performance_tracker.py
```

```
Every time Claude gives analysis with a direction:
Record as a prediction. Check outcome at 24h and 7d.

DATABASE TABLE: predictions
- symbol, prediction_type (bullish/bearish/neutral)
- price_at_prediction, check_24h_at, check_7d_at
- result_24h, result_7d, price_24h_later, price_7d_later

/accuracy → shows bot's prediction track record:
🎯 PREDICTION ACCURACY REPORT
24h Accuracy: 61% | 7d Accuracy: 58%
Best: BTC analysis (73%) | Worst: Altcoins (44%)

---

Create intelligence/memory.py

Build structured trading profile updated weekly:
- preferred_coins, risk_tolerance, trading_style
- known_strengths, known_weaknesses, recent_lessons

Build market insights database:
TABLE: market_insights
- symbol, insight_type, insight_text
- confidence (1-5), times_confirmed, times_denied

Inject personalized context into every Claude call:
"TRADER PROFILE: Swing Trader, Medium Risk, Focuses on BTC/ETH
STRENGTHS: ETH oversold bounces (68% accuracy)
WEAKNESSES: Altcoin selection (44% accuracy)
RECENT LESSON: Wait for volume confirmation before breakouts"

/profile → shows Claude's assessment of your trading style
/insights → shows stored market insights database
/addinsight → manually add a pattern you've noticed
```

LEVEL 17 — Interactive Telegram UI

```
Upgrade bot with Telegram's full interactive features.

INLINE KEYBOARDS FOR ALL COMMANDS:

After /portfolio, add buttons:
```

```
[📊 Analyze Risk] [🔄 Refresh] [➕ Add Position]
[📈 Best Performer] [📉 Worst Performer] [💾 Export]

After /price BTC, add buttons:
[📊 Full TA] [📰 News] [🔍 Research] [🔔 Set Alert]

MAIN MENU (/menu):
[💼 Portfolio] [🔔 Alerts]
[📊 Market]    [📰 News]
[📔 Journal]   [🔍 Research]
[⚙️ Settings]  [❓ Help]

SETTINGS (/settings) as interactive panel:
[🔔 Alert Sensitivity: Medium]
[📅 Briefing Time: 8:00 AM]
[🌙 Night Mode: ON]
[📊 Scanner: Active]

CONFIRMATION DIALOGS for destructive actions:
"⚠️ Delete BTC position? Cannot be undone."
[✅ Yes] [❌ Cancel]

LOADING MESSAGES for long operations:
Edit same message progressively:
"🔍 Starting research..."
→ "📊 Gathering technical data..."
→ "📰 Analyzing news..."
→ "🤖 Claude synthesizing..."
→ [Full report]

PAGINATION for long lists:
── Showing 1-10 of 47 entries ──
[◀️ Previous] [Page 1/5] [Next ▶️]
```

LEVEL 18 — Web Dashboard

```
Add a web dashboard using Flask (port 5000).

PAGES:
/ → Portfolio overview + 7-day chart + market summary
/portfolio → Full positions table + allocation donut chart
/alerts → Active alerts with progress bars to target
/journal → Searchable entry list with filters
/analysis → Enter any coin → get instant analysis
```

```
TECHNOLOGY:
- Flask (lightweight)
- HTML/CSS/JavaScript (no build step)
- Chart.js (CDN) for charts
- Dark theme
- Mobile responsive
- Auto-refresh every 60 seconds

SECURITY:
- Password gate (set in .env: DASHBOARD_PASSWORD=)
- localhost only by default
- /dashboard command in Telegram shows URL + password hint

STARTUP:
Flask runs on port 5000 alongside Telegram bot.
Both start from main.py using threading.

ALL DATA: Read from same SQLite database as bot.
No duplicate logic — pure visualization layer.
```

LEVEL 19 — Workflow Automation

```
Create core/workflow_engine.py

Workflows = sequences of steps that execute automatically.

BUILT-IN WORKFLOWS:

WORKFLOW 1: MORNING PREPARATION (7:45 AM daily)
Steps: warm cache → scan overnight → run skills
→ generate briefing → synthesize with Claude
→ send at exactly 8:00 AM
(Pre-warming ensures instant delivery)

WORKFLOW 2: WEEKLY RESEARCH REFRESH (Sunday 10 AM)
Steps: get portfolio coins → get watchlist coins
→ lightweight research each → update stored insights
→ generate "Week in Review" → send report

WORKFLOW 3: RISK REVIEW (Daily 3 PM)
Steps: calculate portfolio risk score → compare to yesterday
→ if risk increased: identify cause → Claude commentary
→ send alert only if risk elevated
```

```
WORKFLOW 4: OPPORTUNITY SCREEN (9 AM + 9 PM daily)
Steps: run oversold scan → run volume scan → run correlation scan
→ ask Claude to rank by quality → if 2+ high-quality found: send digest

CUSTOM WORKFLOW BUILDER:
/createworkflow → guided setup:
"What triggers this? 1) Time 2) Price event 3) Manual"
"What should it do? List actions:"
Bot creates and schedules workflow.

MONITORING:
/workflows → shows all workflows with last run status and next run time
```

LEVEL 20 — Master Orchestrator

```
Create core/orchestrator.py

Market regime detection (updates hourly):

REGIME 1: BULL TREND (BTC above 50 SMA, F&G >60, 3 green days)
→ Relaxed alerts, momentum-focused scanner, optimistic briefings

REGIME 2: BEAR TREND (BTC below 50 SMA, F&G <40, 3 red days)
→ Increased alerts, risk-reduction focus, cautious briefings

REGIME 3: HIGH VOLATILITY (ATR >2x average)
→ Scanner every 2 min, frequent portfolio checks, volatility warning

REGIME 4: RANGING (tight price range, declining volume, 7+ days)
→ Reduced scanner, breakout setup focus, shorter briefings

ADAPTIVE MESSAGING:
Claude's personality shifts with regime.
Bull: "optimistic but remind about taking profits"
Bear: "emphasize capital preservation"

PRIORITY SYSTEM:
CRITICAL (1): Portfolio -10%+, exchange hack → all channels, no night suppression
HIGH (2): My coin -5%+ in hour, major news → immediate
MEDIUM (3): Scanner finding, opportunity → normal
LOW (4): Weekly summaries, general commentary → scheduled only

COMMANDS:
```

```
/regime → shows current classification with supporting factors
/mode aggressive → lower thresholds for 4 hours then auto-revert
/mode quiet → critical alerts only (good when busy/traveling)
```

PART 3: PROFESSIONAL TIER

Levels 21–30

LEVEL 21 — Quantitative Models + Backtesting

```
Create intelligence/quant_models.py

MODEL 1: MEAN REVERSION CALCULATOR
For any coin/timeframe from 200 periods of Binance data:
- Calculate Z-score = (current price - 50p mean) / std_dev
- Z < -2: Extremely oversold | Z > +2: Extremely overbought
- Historical win rate at current Z-score level
- Expected Value = (Win Rate × Avg Win) - (Loss Rate × Avg Loss)

Output:
MEAN REVERSION ANALYSIS — ETH 4H
Z-Score: -1.87 (Oversold)
Historical edge at this level: 12 instances, 75% win rate
Avg Win: +4.2% | Avg Loss: -2.1%
Expected Value: +2.1% per trade ← Positive EV exists

MODEL 2: MOMENTUM PERSISTENCE
After X% move, does momentum continue or reverse?
Build statistical table: magnitude → continuation probability

MODEL 3: CORRELATION ARBITRAGE
Track rolling 30-day correlations (BTC/ETH, BTC/Gold, ETH/SOL etc.)
When correlation breaks: alert with historical reversion probability

MODEL 4: VOLATILITY REGIME CLASSIFIER
Low/Normal/High/Extreme using ATR + Bollinger width
Historical returns by vol regime

COMMANDS: /edge ETH | /quant | /ev BTC 95000 97000 93000

---
```

```
Create intelligence/backtester.py

Class: Backtester

run_backtest(symbol, strategy, start_date, end_date, capital=10000)

Walks through historical klines chronologically:
- Checks entry conditions each candle
- Simulates fills with slippage
- Tracks portfolio value throughout
- Calculates: total return, Sharpe, Sortino, max drawdown,
  win rate, avg win, avg loss, profit factor

BUILT-IN STRATEGIES TO TEST:
RSI Bounce | Golden Cross | Bollinger Mean Reversion
MACD Crossover | Volume Breakout | Multi-TF Confluence

/backtest ETH RSI 2024-01-01 → run backtest
/strategies → compare all strategies for BTC
/papermode on → start virtual $10k paper trading account
/paper → shows paper trading P&L vs real account
```

LEVEL 22 — DeFi Intelligence Engine

```
Create defi/protocol_monitor.py

MODULE 1: YIELD AGGREGATOR
Pull from DeFiLlama yields API: https://yields.llama.fi/pools
Filter: TVL >$10M, APY >5%, established protocols
Organize by risk: Stablecoin yields | ETH/BTC yields | Altcoin | Leveraged

Daily digest:
💰 DEFI YIELD OPPORTUNITIES
🟢 STABLECOIN: Aave USDC 8.2% ($2.1B TVL), Morpho USDT 11.4%
🟡 ETH: Lido stETH 3.8%, Pendle PT-stETH 7.2% fixed
🔴 HIGH YIELD: Aerodrome WETH/USDC 31.2% (higher risk)

MODULE 2: PROTOCOL HEALTH MONITOR
For each tracked protocol:
- TVL trend (declining = risk signal)
- Revenue vs TVL ratio
- Smart contract audit status
- Admin key risk level
```

```
- Token emission rate (high = APY illusion)

Alert if TVL drops >15% in 7 days.

MODULE 3: IMPERMANENT LOSS CALCULATOR
calculate_il(token_a, token_b, entry_ratio, current_ratio, fees_earned)
"At current fee rate, you break even with IL in 34 days"

---

Create defi/gas_optimizer.py

Track Ethereum gas patterns by hour/day of week.
Best times: Tuesday-Thursday 2-6 AM UTC.

gas_alert_system(): Alert when gas below my threshold.
should_wait_for_lower_gas(tx_value, current_gas):
  If gas >5% of transaction: WAIT with timing recommendation.

/gas → current prices with USD costs for common operations
/gasalert 20 → alert me when gas below 20 gwei
/gasbest → optimal transaction windows from historical patterns
/estimate swap 5000 → cost estimate for specific transaction type
```

LEVEL 23 — Machine Learning Signals

```
Create intelligence/ml_signals.py

pip install numpy==1.26.0 scikit-learn==1.3.0 pandas==2.1.0

MODEL 1: ANOMALY DETECTOR (Isolation Forest)
Features: RSI, volume ratio, price change, Bollinger position, ATR ratio, time of
Trains on 90 days of data. Auto-retrain every Sunday.
Anomaly score -1 to 1 (closer to -1 = more unusual)
Alert when score < -0.6 with explanation of unusual factors.

MODEL 2: REGIME CLASSIFIER (Supervised)
6 classes: Trending Up | Trending Down | Ranging | Breakout Imminent | Distributi
Features: ADX, BB width, volume trend, price position in range, RSI trend
Updates every 4 hours with confidence percentage.

MODEL 3: PRICE TARGET PREDICTOR (Gradient Boosting)
Predicts likely price range at 24h, 48h, 7d.
Outputs: pessimistic/base/optimistic range with confidence.
```

```
"Model accuracy last 30 days: 61% within range"

TRANSPARENCY RULE:
Always show top 3 features driving each prediction.
Never show output without showing reasoning.

COMMANDS:
/ml BTC → all model outputs for Bitcoin
/anomaly → anomaly scores for all portfolio coins
/forecast ETH → price forecast
/regime → current regime for all portfolio coins
```

LEVEL 24 — Social Intelligence

```
Create data/social_intelligence.py

FREE DATA SOURCES:
1. Reddit API (free, no key): r/cryptocurrency, r/bitcoin, r/ethereum hot/new
2. LunarCrush API (free tier): social volume + galaxy score
3. CryptoPanic vote ratios (already in news module)
4. pytrends: Google Trends for "Bitcoin", "buy Bitcoin", "crypto crash"

FOMO/PANIC DETECTOR:
FOMO signals (mark local tops):
"Should I sell my car to buy BTC?" | "Going all in" | "Never felt more confident"

PANIC signals (mark local bottoms):
"Just sold everything" | "Crypto is dead" | "Lost my savings"
Mainstream news: "Crypto crash"

Alert when either extreme detected.

COMPOSITE SOCIAL SCORE:
SOCIAL SENTIMENT DASHBOARD
Overall Score: 7.2/10 (Bullish) ↑ Rising
Reddit: 2,847 posts (+34%) | Sentiment: 68% positive
LunarCrush: 4.2x above average volume 🔥
Google: "Buy Bitcoin" searches +180% this week
⚠️ CONTRARIAN WARNING: Extreme retail FOMO detected

SCHEDULED: Every 6 hours. Alert when score changes >20%.
COMMANDS: /social | /social BTC | /fomo
```

LEVEL 25 — Institutional Flow Tracker

```
Create intelligence/institutional_tracker.py

FREE DATA SOURCES:
1. SEC EDGAR: 13F filings mentioning "bitcoin"
   https://efts.sec.gov/LATEST/search-index?q=%22bitcoin%22&forms=13F
2. Farside ETF flows (public scrape): farside.co.uk/bitcoin-etf-flow-all-data-ser
3. CFTC COT Report (weekly): cftc.gov/dea/futures/deacmesf.htm
4. Grayscale trust premium/discount (public)

get_etf_flows():
🏦 US SPOT BTC ETF FLOWS
BlackRock IBIT: +$312M | Fidelity FBTC: +$89M
NET TODAY: +$423M | 7-Day Net: +$1.2B (strong institutional buying)

analyze_futures_positioning():
"Large speculators NET LONG 45,000 contracts (85th percentile).
Extreme positioning can precede corrections."

track_sec_filings(): Alert on new 13F positions.

WEEKLY INSTITUTIONAL REPORT (Sunday):
ETF net flows, futures positioning change, notable 13F activity, Grayscale premiu

COMMANDS: /institutional | /etfflows | /cot
SCHEDULED: ETF flows daily 6 PM, COT every Saturday.
```

LEVEL 26 — Multi-Chain Monitor

```
Create blockchain/cross_chain_monitor.py

CHAINS: Ethereum, Solana, Bitcoin, Arbitrum, Base, Avalanche, Polygon

FREE APIs:
- Etherscan/Arbiscan/Basescan (same API format, free keys)
- Helius free tier for Solana
- Blockchain.info + Mempool.space for Bitcoin
- DeFiLlama for cross-chain TVL

get_chain_health(chain): TPS, active addresses, avg fee, health score 1-10

compare_chain_activity():
CHAIN ACTIVITY COMPARISON (24h):
```

```
Chain     TPS    Addresses  Avg Fee  TVL
Ethereum  25     892,000    $8.20    $67B
Solana    3,200  1,240,000  $0.001   $12B
Base      45     567,000    $0.08    $2.1B

track_bridge_flows(): Monitor Stargate/Across/Hop
"$45M USDC bridged ETH→Base (last 4h). DEX volumes up 23%."

bitcoin_network_health(): Hash rate, difficulty adjustment, mempool, Lightning ca

Daily cross-chain briefing injected into morning report.

COMMANDS: /chains | /chain ETH | /chain SOL | /bridges
```

LEVEL 27 — Voice Interface

```
Create bot/voice_handler.py

pip install openai  (or use Groq Whisper — free tier, faster)
pip install gTTS

GROQ WHISPER (free):
https://api.groq.com/openai/v1/audio/transcriptions
Model: whisper-large-v3-turbo

When voice message received:
1. Download from Telegram
2. Transcribe with Groq Whisper
3. Show: "🎙 I heard: '[transcription]'"
4. Process as normal text through full pipeline
5. Convert response to audio with gTTS
6. Send back as voice message + text version

VOICE-OPTIMIZED PROMPT:
Add when voice detected:
"This will be read aloud. Use natural speech:
- No markdown, symbols, or formatting
- Short sentences, conversational tone
- Spell out numbers ('ninety-seven thousand')
- Under 150 words for comfortable listening"

CHART PHOTO HANDLING:
When photo received, use Claude vision:
"Describe what you see in this crypto chart
```

```
and provide technical analysis."

COMMANDS:
/voice on → enables voice responses
/voice off → text only (default, faster)
```

LEVEL 28 — Kelly Criterion & Position Sizing

```
Create portfolio/position_sizer.py

MODEL 1: KELLY CRITERION
K = W - (1-W)/R
Where W = win rate (from predictions table), R = avg win / avg loss

Calculate from actual trading history.
Recommend Half Kelly (safer).

Output:
KELLY CRITERION ANALYSIS (last 50 trades)
Win Rate: 62% | Avg Win: +7.3% | Avg Loss: -3.8%
Full Kelly: 25.8% | Half Kelly (recommended): 12.9%
For $50k portfolio: optimal position = $6,450 per trade

MODEL 2: VOLATILITY-ADJUSTED SIZING
Size = Risk_Amount / (ATR × Multiplier)
/size BTC 500 95000 → calculates exact position for $500 risk at $95k entry
Shows: quantity, stop loss, take profit targets, R:R ratio

MODEL 3: PORTFOLIO HEAT
Total % at risk across ALL open positions simultaneously.
Max heat: 15% | Max single position: 5%
/heat → "Total Heat: 5.3% ✅ (9.7% capacity remaining)"

MODEL 4: DRAWDOWN PROTECTION
After 3 consecutive losses → reduce size 25%
After 5 consecutive losses → reduce size 50%
After portfolio -10% in month → reduce size 75%
Auto-enforced with alerts when rules activate.

/sizing → full dashboard with all models
Auto-appended to /analyze and trade discussions.
```

LEVEL 29 — Options Intelligence

```
Create derivatives/options_monitor.py

FREE DATA: Deribit public API (no auth needed):
https://www.deribit.com/api/v2/public/get_book_summary_by_currency?currency=BTC&k

KEY METRICS:

PUT/CALL RATIO:
<0.7: Very bullish | 0.7-0.9: Mildly bullish
1.1-1.3: Mildly bearish | >1.3: Very bearish
Contrarian: >1.5 often marks bottoms, <0.5 often marks tops

MAX PAIN: Strike where most options expire worthless.
"Price gravitates toward max pain near expiry."

IMPLIED VOLATILITY SURFACE:
Current IV vs 30-day historical vol.
If IV > HV: options expensive (potential to sell)
If IV < HV: options cheap (potential to buy)
Term structure: short-term vs long-term IV

GAMMA EXPOSURE:
Positive GEX → price moves amplified (more volatile)
Negative GEX → price moves dampened
Key levels: "Gamma flip at $95k — if price falls below, expect acceleration"

UNUSUAL ACTIVITY DETECTOR:
Single large option purchases (>$5M)
Sudden IV spikes at specific strikes
"🐳 UNUSUAL OPTIONS ACTIVITY: $24M BTC calls purchased at $110k strike, 35 days e

SCHEDULED: Pull every 4 hours. Alert on unusual activity immediately.
Include put/call ratio in morning briefing.

COMMANDS: /options | /options SOL | /maxpain | /iv
```

LEVEL 30 — Multi-Analyst Debate System

```
Create intelligence/debate_system.py

THREE ANALYST PERSONAS:
```

```
🐂 BULL (Momentum Seeker):
System prompt: "You are an aggressive momentum trader.
Find: strong uptrends, high-volume breakouts, RSI divergences.
Present the strongest possible bull case in 100 words."

🐻 BEAR (Risk Manager):
System prompt: "You are a conservative risk manager.
Find: overbought conditions, resistance, macro headwinds, distribution.
Present the strongest possible bear case in 100 words."

📊 QUANT (Data Scientist):
System prompt: "You are a quantitative analyst who only trusts numbers.
Calculate: historical win rates, expected value, statistical edge.
Present what the data shows with confidence levels."

DEBATE PROCESS:
Round 1: All three analyze identical data. 100-word opening each.
Round 2: Bull rebuts Bear. Bear rebuts Bull. Quant evaluates both.
Round 3: Moderator Claude synthesizes:
- Where do they agree? (high confidence)
- Where do they disagree? (uncertainty)
- Most likely outcome?
- Recommended action with reasoning.

COMMANDS:
/debate ETH → full 3-analyst debate about ETH
/debate "should I sell my BTC at $100k" → specific question
/quickdebate BTC → opening statements + synthesis only (60 seconds)
```

PART 4: ELITE TIER

Levels 31–40

LEVEL 31 — Event Impact Predictor

```
Create intelligence/event_predictor.py

TRACKED EVENT TYPES:

1. TOKEN UNLOCKS (from token.unlocks.app or cryptorank.io):
For each upcoming unlock:
```

```
- Calculate % of circulating supply unlocking
- Analyze historical price action ±30 days around previous unlocks
- Model likely impact

Output:
⚠️ TOKEN UNLOCK ALERT
ARB — 1.1B ARB ($890M) | Unlock: March 15 (19 days)
Historical: avg -12% in week before, -8% unlock day, +5% 30 days after
Prediction: Selling pressure next 2 weeks likely (73% historical base rate)

2. PROTOCOL UPGRADES:
Track Ethereum hard forks, major DeFi upgrades.
Historical: "3/5 last Ethereum upgrades saw price increase 2 weeks after.
Pattern: buy the rumor, sell the news"

3. MACRO EVENTS:
Store historical crypto response to Fed meetings, CPI, jobs:
"HAWKISH SURPRISE: avg -8.2% BTC in 24h (high confidence)"
"FED IN-LINE: avg +0.5%"

4. OPTIONS/FUTURES EXPIRY:
Monthly options expiry (last Friday):
Historical: elevated volatility week before, max pain gravity.

DATABASE: upcoming_events table with predicted impact and confidence.
Alert 7 days and 1 day before each event.

/calendar → 30-day event calendar with color-coded urgency
/predict ARB unlock → full unlock impact analysis
```

LEVEL 32 — Macro Correlation Engine

```
Create intelligence/macro_monitor.py

FREE DATA:
FRED API (free key at fred.stlouisfed.org): CPI, Fed funds rate, yield curve
yfinance: SPX, Gold (GLD), Bonds (TLT), DXY, VIX, Oil

pip install yfinance

fetch_traditional_market_data():
S&P 500, Gold, DXY (USD index), VIX (fear gauge), 10Y yield, WTI Oil

calculate_correlations(lookback=90):
```

```
BTC/SPX, BTC/Gold, BTC/DXY (usually inverse), BTC/VIX (usually inverse)
Track how correlations change: high correlation = crypto follows macro.

macro_regime_detector():
RISK ON: SPX rising, VIX <20, USD neutral → Crypto bullish
RISK OFF: SPX falling, VIX >30, USD strengthening → Crypto bearish
LIQUIDITY EXPANSION: Fed balance sheet growing → historically very bullish BTC

dollar_cycle_analysis():
DXY cycles are major crypto drivers.
"BTC outperforms when DXY falls below 50-week MA.
Current DXY: -2.3% vs 50w MA → Bullish signal"

Inject macro context into morning briefing.
Alert when macro regime changes.

COMMANDS: /macro | /correlation | /dxy
```

LEVEL 33 — Advanced Task Queue

```
Create core/task_queue.py

PROBLEM: Morning briefing needs 10 data sources simultaneously.
Doing them sequentially = slow and brittle.

SOLUTION: Async parallel execution with dependency graph.

TASK DEFINITION:
{
  'name': 'fetch_btc_price',
  'function': price_service.get_price,
  'args': ['BTC'],
  'priority': 1-10,
  'depends_on': [],
  'timeout': 10,
  'retry_count': 3,
  'retry_delay': 2,
}

EXECUTION:
1. Identify tasks with no dependencies → run in parallel
2. As each completes → unlock dependent tasks
3. Run newly unlocked in parallel
4. Repeat until all complete
```

```
CIRCUIT BREAKER:
If an API fails 5 times in 10 minutes:
- Open the circuit for that API for 5 minutes
- Use fallback if available
- Log circuit breaker activation

MORNING BRIEFING OPTIMIZED:
All 5 data fetches run simultaneously.
Total time: ~3 seconds (vs 15+ sequential).

RETRY LOGIC:
Exponential backoff: delay doubles each retry.
All retries fail → call on_failure fallback.

/taskstatus → queue status, circuit breakers, slowest tasks today
```

LEVEL 34 — Advanced Security Hardening

```
Create security/security_manager.py

LAYER 1: ENHANCED AUTHENTICATION
- Rate limiting: 30 messages/min, 100 commands/hour
- Session timeout: re-auth after 12 hours inactivity
- 2FA for sensitive commands: /backup, /clearall
- PIN stored hashed. 3 wrong attempts = 30-min lockout.

LAYER 2: DATA ENCRYPTION
pip install cryptography
Encrypt with Fernet: journal entries, trade positions, wallet addresses.
Key derived from Telegram ID + environment secret.
Transparent: encrypt on write, decrypt on read.

LAYER 3: AUDIT LOGGING
Every action logged: who, what command, what data accessed, timestamp, result.
Logs: encrypted, append-only, cannot be modified.
/auditlog → last 50 actions + security events

LAYER 4: SENSITIVE DATA AUTO-DELETE
Portfolio messages auto-delete after 60 seconds.
"(auto-deleting in 60 seconds)" notice appended.
Uses Telegram delete_message API with asyncio.sleep.

LAYER 5: ANOMALY DETECTION
```

```
Commands at unusual hours? Unusual volume of sensitive requests?
"⚠️ SECURITY NOTICE: 5 commands sent 3-4 AM. Was this you?"
[Yes, it was me] [No, investigate]

LAYER 6: DATA MINIMIZATION
Auto-cleanup weekly:
Conversation history: keep 30 days
Scanner logs: keep 7 days
Price cache: keep 1 hour
Old completed alerts: keep 90 days

/security → full security status dashboard
```

LEVEL 35 — Performance Attribution

```
Create analytics/performance_attribution.py

DATABASE: portfolio_snapshots (daily) + return_attribution

ATTRIBUTION BREAKDOWN:

1. ASSET SELECTION: Did you pick the right coins?
   "ETH returned +23.1% vs BTC +31.2%. ETH cost you 8.1% relative."

2. ALLOCATION EFFECT: Did you size correctly?
   "20% SOL with SOL +45% contributed +9% to portfolio."

3. TIMING EFFECT: When did you add/reduce?
   "You added ETH at avg $3,050 vs period avg $3,200. Timing advantage: +4.9%"

4. BENCHMARK COMPARISON:
   vs BTC buy-and-hold: your alpha = +X%
   vs Equal-weight top 10: your alpha = +Y%

5. FACTOR ANALYSIS:
   Market beta drove X% | Sector drove Y% | Your skill drove Z%
   "Skill = what you actually controlled vs market exposure"

MONTHLY WATERFALL CHART DATA:
Components that add/subtract from total return.
Visualized in web dashboard.

/performance → comprehensive attribution report
/attribution 2026-01-01 2026-02-23 → specific date range
```

```
/compare → compare two periods side by side
```

LEVEL 36 — Autonomous Trade Proposals

```
Create intelligence/trade_proposer.py

# DISCLAIMER: Only PROPOSES trades for human review.
# Never executes automatically.
# All proposals require explicit human approval.

SETUP DETECTORS (all conditions must be true):

SETUP 1: OVERSOLD QUALITY BOUNCE (8/10 conviction)
RSI <35 on 4h | Above 200-day SMA | Volume declining | Top-30 market cap

SETUP 2: MOMENTUM BREAKOUT (7/10)
Breaks 30-day high with 2x+ volume | ADX rising >25 | RSI 50-65

SETUP 3: FUNDING RATE RESET LONG (6/10)
Funding was negative 24h | Now normalizing | OI declining

SETUP 4: MACRO DIVERGENCE (6/10)
Risk-on macro | Coin underperforming sector 10%+ | No fundamental reason

SETUP 5: EVENT-DRIVEN (7/10)
Major upgrade in 2-4 weeks | Not already priced in | Historical runs confirmed

PROPOSAL FORMAT:
💡 TRADE PROPOSAL — ETH OVERSOLD BOUNCE (8/10)
RSI: 32.1 | Above 200 SMA: ✅ | Volume: Declining ✅
Entry: $3,050-$3,150 | Stop: $2,950 | Target 1: $3,350 (2.7:1 R/R)
Position: 0.16 ETH ($496 risk from $50k portfolio)
Historical win rate at this setup: 68% (EV: +4.7%)
Bull/Bear/Quant consensus: [one line each]
[✅ Add to watchlist] [📔 Journal] [🔔 Set Alert] [❌ Dismiss]

/proposals → all active proposals ranked by conviction
Track accepted/rejected proposals + outcomes for self-improvement.
```

LEVEL 37 — REST API + TradingView Integration

```
Create api/server.py

pip install flask flask-restful flask-limiter

AUTH: X-API-Key header. Random 32-char key in .env.
RATE LIMIT: 100 requests/hour per key.

ENDPOINTS:
GET  /api/portfolio → positions with live prices + P&L
GET  /api/prices?symbols=BTC,ETH,SOL → current prices
GET  /api/alerts → all active alerts with distances
POST /api/alerts → create new alert
DELETE /api/alerts/{id} → cancel alert
GET  /api/market → market overview data
GET  /api/technical/{symbol}?timeframe=4h → TA data
POST /api/journal → add entry remotely
GET  /api/signal/{symbol} → composite signal for coin
GET  /api/stats → performance statistics

POST /api/tradingview/webhook
Receives TradingView Pine Script alerts:
Body: {"symbol": "BTCUSDT", "action": "buy_signal",
       "price": "{{close}}", "message": "RSI oversold on 4h"}
→ Creates journal entry + sends Telegram notification

TRADINGVIEW SETUP IN CODE COMMENTS:
"""
1. Create alert in TradingView
2. Webhook URL: http://your-server:8080/api/tradingview/webhook
3. Message: {"symbol": "{{ticker}}", "price": "{{close}}", "message": "{{strategy
"""

/api command in Telegram shows URL, key, and all endpoints.
```

LEVEL 38 — Simulation Environment

```
Create simulation/sandbox.py

Full sandbox for testing without real API calls or database changes.

FAKE DATA GENERATORS:
generate_fake_portfolio() → realistic test portfolio
generate_fake_market_data() → realistic prices with crypto volatility
```

```
generate_fake_price_history(symbol, days) → synthetic OHLCV
generate_fake_news() → structured fake news items

MOCK SERVICES:
MockTelegramBot → stores messages in list instead of sending
MockAnthropicClient → returns pre-written responses (no API cost)
MockExchangeAPI → fake prices + simulated failures

SIMULATION SCENARIOS:

SCENARIO 1: MARKET CRASH
BTC drops 20% over 4 hours. Verify:
✓ Alert system fires | ✓ Portfolio monitor detects | ✓ No duplicate alerts

SCENARIO 2: API FAILURE CASCADE
All price APIs go offline. Verify:
✓ Fallback to cache | ✓ Friendly error messages | ✓ Circuit breakers open

SCENARIO 3: HIGH VOLUME
1000 rapid messages. Verify:
✓ Rate limiting works | ✓ Queue processes correctly | ✓ Bot stays stable

TEST RUNNER: /test
Runs all tests and shows:
31/31 tests passed ✅ | Failed: 0 | Time: 4.2s

/simulate crash → shows what alerts would fire (nothing sent to Telegram)
/simulate spike 15 → shows system response to 15% pump
```

LEVEL 39 — Personal Crypto Academy

```
Create education/knowledge_base.py

LEARNING PATHS:

PATH 1: TECHNICAL ANALYSIS (10 lessons)
1. What is TA and why it works/doesn't
2. Support and resistance — the foundation
3. Trend identification (HH/HL/LH/LL)
4. Volume interpretation
5. RSI — momentum oscillator
6. Moving averages — trend following
7. MACD — momentum + trend
8. Bollinger Bands — volatility
```

```
9. Fibonacci retracements
10. Multi-timeframe analysis

PATH 2: DEFI MASTERY (10 lessons)
Liquidity pools, IL, yield farming, lending, stablecoins,
bridges, smart contract risks, governance, MEV

PATH 3: RISK MANAGEMENT (10 lessons)
Position sizing, Kelly Criterion, portfolio correlation,
stop losses, R:R ratios, drawdown psychology, tax implications

ADAPTIVE LEARNING:
Track which topics you've asked about.
After asking about RSI 3 times: assume you know it.
Flag knowledge gaps from trading performance.
"Your funding rate signals have 58% accuracy. Suggested lesson: Funding Rate Trad

DAILY CONCEPT (in morning briefing):
📚 CONCEPT OF THE DAY: [term + 3-sentence explanation + link to full lesson]

QUIZ SYSTEM:
/quiz TA → 5-question multiple choice quiz
Track weak areas. Suggest review for lowest scores.

COMMANDS:
/learn → dashboard with progress bars
/learn rsi → deep dive lesson with examples
/explain "impermanent loss" → on-demand explanation
/quiz → quiz on any topic
```

LEVEL 40 — Unified Intelligence Hub

```
Create core/intelligence_hub.py

SIGNAL AGGREGATION:
Every hour, collect from all modules:
- Technical score (-3 to +3) per coin
- ML anomaly + regime signal
- Social sentiment score
- Macro regime score
- Institutional flow direction
- On-chain signal
- Quant models EV
```

```
WEIGHTED COMPOSITE SCORE (-10 to +10):
Weight each signal by historical accuracy + signal strength + recency.

🧠 UNIFIED MARKET INTELLIGENCE
BTC: +6.2/10 🟢 STRONGLY BULLISH
ETH: +2.1/10 🟡 MILDLY BULLISH
SOL: +4.8/10 🟢 BULLISH

Signal breakdown shows contribution of each source.

DAILY ACTION AGENDA:
Every morning, synthesize into prioritized agenda:
PRIORITY 1 — MONITOR: BTC approaching $100k resistance
PRIORITY 2 — RESEARCH: SOL unusual social volume
PRIORITY 3 — REVIEW: ETH position 18 days old, thesis played out
PRIORITY 4 — LEARN: Your funding signals need improvement
NO ACTION: BNB fine, all alerts properly set

WEEKLY INTELLIGENCE REPORT (Sunday):
Regime analysis | Signal accuracy review | Portfolio attribution
Top 3 opportunities next week | Risks to monitor | Action plan

COMMANDS:
/hub → unified intelligence dashboard
/agenda → today's prioritized action agenda
/signals → all individual signals feeding into composite score
/weekly → trigger weekly intelligence report
/intelligence → the ONE most important thing to know right now
```

PART 5: FRONTIER

Tiers 5–10: Beyond the System

THE AUDIT PROMPT (Do This Before Adding More)

```
I've built a complex crypto trading system with 40 modules.
Before adding anything new, help me audit what I actually use.

Here are my database stats:
- Commands sent last 30 days: [paste /stats output]
- Most used commands: [paste list]
```

```
- Least used commands: [paste list]
- API costs last month: [paste amount]

Questions I need answered:
1. Which modules am I clearly NOT using? (less than 5 times in 30 days)
2. Which modules cost most in API calls relative to their value?
3. What is the SIMPLEST version of this system that captures 90% of value?
4. What ONE thing, if done consistently every day, would improve my trading
   more than any new feature?

Be brutally honest. I would rather have a smaller system I actually use
than a large system I'm impressed by but don't use.
```

TIER 5 — Alternative Data Pipeline

```
Create data/alternative_data.py

SIGNAL 1: JOB POSTING INTELLIGENCE
Parse LinkedIn/Indeed/Greenhouse for crypto company job postings.
Signal: Sudden 15+ engineering jobs = product launch coming.
"Restructuring advisor" hiring = trouble.
Tool: Apify or Playwright (free tiers available).

SIGNAL 2: GITHUB COMMIT VELOCITY
Track commit frequency on key protocol repos:
https://api.github.com/repos/{owner}/{repo}/commits
Alert: Development spike often precedes announcements.
Sudden silence often precedes bad news.

SIGNAL 3: SEC EDGAR MONITORING
Search for crypto mentions before they hit news sites:
https://efts.sec.gov/LATEST/search-index?q=bitcoin&forms=litreleases
Typically 2-6 hours ahead of crypto news sites.

SIGNAL 4: DERIVATIVES FLOW INTELLIGENCE
Deribit real-time large block trades.
$5M+ in out-of-money calls = someone knows something or is very wrong.
Track: unusual options blocks with timing vs price outcome.

SIGNAL 5: APP STORE RANKINGS
Coinbase, Binance, Kraken rankings.
Rising rank = new retail money incoming.
Free: scrape App Store search position or SensorTower free tier.
```

TIER 6 — Cross-Market Arbitrage Intelligence

```
Create intelligence/arbitrage_scanner.py

INFORMATION ARBITRAGE (not trade execution):

SCANNER 1: FUTURES BASIS TRACKER
ETH Spot (Coinbase): $3,100 | ETH Perp (Binance): $3,094
Basis: -0.19% | Annualized: -28%
When basis extremely negative: longs collect funding.

SCANNER 2: CROSS-EXCHANGE PREMIUM
Same coin, different prices:
BTC Coinbase: $97,540 | Kraken: $97,480 | Binance: $97,510
Persistent Coinbase premium = US institutional buying pressure.

SCANNER 3: DEFI vs CEX PRICING
Large Uniswap vs Binance gap = arbitrage closing soon.
Watch the gap closing: signals direction of next move.

SCANNER 4: STABLECOIN PEG MONITOR
USDT $0.997 = fear signal.
USDT premium in certain regions = capital flight indicator.
During March 2023 USDC depeg: those monitoring had 6-hour warning.
```

TIER 7 — Behavioral Pattern Analyzer

```
Create psychology/behavior_analyzer.py

ANALYSIS 1: TIME-OF-DAY PERFORMANCE
Query journal: group wins/losses by hour of day.

YOUR PERFORMANCE BY HOUR:
6-9 AM: 71% win rate ✅ | 9-12: 65% ✅ | 12-2 PM: 58% ⚠️
5-8 PM: 44% ❌ | 8 PM+: 38% ❌ (FOMO trading)

"Your worst trading is after 5 PM.
A no-new-positions-after-5PM rule would have saved ~$1,200 last 3 months."

ANALYSIS 2: POST-LOSS BEHAVIOR
What happens in your next 3 trades after a loss?
"After a loss, your next trade is larger (67% of the time) — revenge trading.
```

```
After 2 consecutive losses: win rate drops to 31%, position size increases 43%."

ANALYSIS 3: SETUP QUALITY vs POSITION SIZE
Does size correlate with conviction? It should.
"Your position sizing has 0.12 correlation with setup quality. Essentially random
Better sizing could add 15-20% to annual returns."

BEHAVIORAL RULES ENGINE:
Based on YOUR data, suggest rules:
"Data suggests: no trading after 5 PM, 2-hour break after any loss.
Want me to add these as enforced reminders?"

WEEKLY PSYCHOLOGY REPORT alongside performance report. No judgment — just data.
```

TIER 8 — Self-Improving Prompt System

```
Create intelligence/prompt_optimizer.py

CONCEPT:
Every Claude prompt can be measured and improved automatically.

MEASUREMENT:
- Thumbs up/down per response → explicit feedback
- Did user do anything with this response? → action rate
- Was the prediction accurate? → outcome rate
- How long did user engage? → depth indicator

A/B TESTING:
Version A: Current morning briefing prompt.
Version B: Slightly modified version.
After 14 days: keep whichever performs better.
Generate Version C. Repeat.

TRACKING DATABASE:
TABLE: prompt_versions
- prompt_name, version, prompt_text, created_at
- helpfulness_score, prediction_accuracy, action_rate
- engagement_depth, times_used, is_active

MONTHLY ANALYSIS:
"Your morning briefing prompt has improved 23% in user satisfaction
over 3 versions in 90 days. Current best version is V4.
Testing V5 now with modified risk framing."
```

```
After 6 months: Your prompts are dramatically better than any default,
tuned to how your brain works and what you actually act on.
```

TIER 9 — Infrastructure Decisions

```
HONEST ROI ANALYSIS FOR EACH UPGRADE:

ETHEREUM ARCHIVE NODE ($40-50/month VPS, 3TB NVMe):
Software: Reth (Rust, fastest sync)
Savings vs: Alchemy/Infura at $200+/month for heavy usage
Gains: Unlimited RPC calls, any historical state, 50-100ms vs 300-500ms latency
Verdict: Pays for itself if spending $200+/month on RPCs.

SOLANA RPC NODE (~$100/month):
Gains: Jito MEV access, fast submission, Jupiter without rate limits
Verdict: Only if doing serious Solana DeFi.

FLASHBOTS PROTECT (free):
Private transaction submission. No front-running.
For any DeFi transaction over $5,000: saves money immediately.

CO-LOCATION (exchanges allow it):
Latency: 0.3ms vs 40-150ms retail.
Only matters for time-sensitive arbitrage. Most people don't need this.

TIMESCALEDB: For time-series market data (vs SQLite)
REDIS: Sub-millisecond price cache
KAFKA: Event streaming if building distributed system

Recommendation: Start with archive node when RPC costs exceed $40/month.
Add Redis when SQLite cache becomes a bottleneck.
Everything else: only when you have a specific problem it solves.
```

TIER 10 — The Final Form (What It Becomes After 1 Year)

```
After 12 months of daily use, the system has:

ACCUMULATED DATA:
12 months trade history | 12 months annotated market data
Thousands of predictions with outcomes | Your behavioral patterns at every condit
```

ACCUMULATED INTELLIGENCE:
Claude knows your trading style better than any human advisor
ML models trained on YOUR specific history
Prompts tuned to how you think and act
Rules built from your actual behavioral data

THE COMPOUND EFFECT:
Month 1: Useful tool
Month 3: Helpful advisor that knows you
Month 6: Recognizes your patterns before you do
Month 12: Better than any human advisor at knowing YOUR situation
Month 24: The most valuable financial tool you own

THE THREE THINGS THAT ACTUALLY MATTER:

1. CONSISTENCY beats sophistication.
   A simple system used daily beats a complex one used occasionally.

2. THE JOURNAL is the most valuable thing you built.
   Not the ML models. Not the options monitor.
   Six months of honest journaling + performance tracking
   will change your trading more than any indicator.

3. THE SYSTEM IS A MIRROR, NOT A CRUTCH.
   The goal was never to remove your judgment.
   It was to give your judgment better inputs
   and hold it accountable with honest data.

THE FINAL PROMPT:
"Given everything I've built, given my actual usage patterns,
given my performance data — what is the single highest-leverage
thing I could do with this system in the next 30 days
to actually improve my trading results?

Not a new feature. Not a new integration.
The actual highest-leverage action.

Here is everything you know about me:
[paste /profile output]
[paste /accuracy output]
[paste /performance output]
[paste /behavior output]

Give me one honest answer."

PART 6: DASHBOARD & DISCORD

Visual Interface + Multi-Platform

PROMPT: FULL WEB DASHBOARD

```
I have a crypto intelligence agent built in Python with a Telegram bot,
SQLite database, and multiple intelligence modules.

Build a COMPLETE web dashboard that visualizes everything in real-time.

TECH STACK:
- Backend: FastAPI (async support)
- Frontend: Single HTML file
  * TailwindCSS (CDN)
  * Chart.js (CDN)
  * Alpine.js (CDN) for reactivity
- Real-time: Server-Sent Events (SSE)
- Database: Read directly from existing SQLite

DESIGN SYSTEM:
Background: #0a0a0f | Cards: #12121a
Accent: #6366f1 (indigo) | Gain: #10b981 | Loss: #ef4444
Warning: #f59e0b | Text: #e2e8f0 | Font: Inter (Google Fonts)

═══════════ PAGE 1: COMMAND CENTER (/) ═══════════

TOP BAR: Agent status + time + quick stats row:
[Portfolio Value] [24h Change] [BTC Price] [ETH Price] [F&G] [Active Alerts]

LEFT SIDEBAR: Navigation icons
🏠 Command Center | 💼 Portfolio | 📡 Scanner | 🔔 Alerts
📔 Journal | 📊 Analytics | 🤖 AI Chat | ⛓️ On-Chain | 📰 News | ⚙️ Settings

MAIN GRID (3 columns desktop):
Col 1: Portfolio donut chart + Market heatmap (top 20 coins, color = magnitude)
Col 2: Portfolio value line chart (30D default, 7D/90D/ALL toggle, vs BTC benchma
       + Unified Intelligence Score gauge (-10 to +10) with signal breakdown tabl
Col 3: Live activity feed (real-time scrolling events, color-coded by type)
       + Agent status panel (all background services with status/last run/next ru

═══════════ PAGE 2: PORTFOLIO (/portfolio) ═══════════
```

Header: Total value, 24h/7d/30d change, all-time P&L

Tabs: [Positions] [Performance] [Risk] [History]

POSITIONS tab: Live-updating table with coin logos
PERFORMANCE tab: Equity curve + Sharpe/Sortino/Max DD stats + monthly heatmap cal
RISK tab: 4 gauges (Concentration/Heat/Correlation/Drawdown) + sizing recommendat
HISTORY tab: All closed trades, sortable, filterable, CSV export

========== PAGE 3: SCANNER (/scanner) ==========

5 scan type toggle cards at top.
Left: findings list (last 24h) with severity badges.
Right: Selected finding detail + mini chart + Claude analysis + action buttons.

========== PAGE 4: ALERTS (/alerts) ==========

Add alert form (collapsible). Active alerts as cards with:
- Distance progress bar (fills as price approaches target)
- Glowing border when within 5% of target
- [Edit] [Delete] [Pause] per card

========== PAGE 5: JOURNAL (/journal) ==========

Left: filter panel (search, date range, coin, type, sentiment, outcome)
Main: entry cards (expand on click) + quick-add entry at bottom
Analytics sidebar: consistency calendar + sentiment chart

========== PAGE 6: ANALYTICS (/analytics) ==========

Tabs: [Performance] [Predictions] [Psychology] [Backtests] [Attribution]

PSYCHOLOGY tab: Win rate by hour chart, by day chart, post-loss behavior,
position sizing scatter plot, key behavioral insights.

========== PAGE 7: AI CHAT (/chat) ==========

Left panel: What Claude currently knows (portfolio/market/signals checklist)
Main: Chat bubbles with code highlighting + table rendering + chart embedding
Input: Text area + voice button + quick prompts above:
[Market Overview] [My Portfolio?] [Best Setup Now?] [Weekly Review]

========== PAGE 8: ON-CHAIN (/onchain) ==========

Chain selector tabs. Per chain: key metric cards, gas/flow charts,
large transaction live feed with etherscan links.

```
========== PAGE 9: NEWS (/news) ==========

Sentiment bar at top. News cards with source, sentiment badge, impact score.
Sidebar: sentiment over time chart.

========== PAGE 10: SETTINGS (/settings) ==========

Sections: Agent Config | Notification Channels | Portfolio Settings
| API Connections (status indicators) | Display Preferences | Data Management

========== GLOBAL FEATURES ==========

SSE endpoint /api/stream — streams price updates (30s), scanner alerts (immediate
activity events (immediate), portfolio recalculation (60s).

Keyboard shortcuts: G+C/P/A/J/N for navigation, / for AI chat, ? for help.

SECURITY: Password gate in .env: DASHBOARD_PASSWORD=
Session cookie, 24-hour expiry.

FILE STRUCTURE:
dashboard/app.py (FastAPI + all routes)
dashboard/index.html (complete single-page frontend)

Here is my current codebase: [PASTE ALL YOUR FILES]
```

PROMPT: DISCORD AGENT

```
Add a full Discord integration. The agent should operate natively
in Discord with its own identity, not just forward Telegram messages.

pip install discord.py

========== SETUP GUIDE (include as comments) ==========
"""
1. discord.com/developers/applications → New Application → "CryptoAgent"
2. Bot tab → Add Bot → Copy TOKEN → .env DISCORD_TOKEN=
3. Enable: Message Content Intent, Server Members Intent, Presence Intent
4. OAuth2 → URL Generator → scopes: bot, applications.commands
5. Permissions: Send/Read Messages, Create Threads, Embed Links,
   Add Reactions, Manage Messages
6. Create server channels:
   #crypto-agent #price-alerts #market-scanner
   #morning-briefing #journal #signals
```

```
   Copy each channel ID → .env
"""

═══════════ DISCORD PERSONALITY ═══════════

Slightly more casual than Telegram. Use Discord markdown (**bold**, `code`).
Use embeds for all data. Suggest threads for long content.
Never share exact portfolio values in public channels.

═══════════ SLASH COMMANDS ═══════════

/price [coin] → price embed with 24h change
/market → full market overview embed
/ta [coin] [timeframe] → technical analysis embed
/portfolio → EPHEMERAL (only visible to requester)
/alert [coin] [above/below] [price] → creates alert
/research [coin] → announces start, posts full report as thread
/signals → high-conviction opportunities (posts to #signals)
/briefing → on-demand briefing with @here
/journal [entry] → logs trade, confirms with ✅
/debate [question] → three-analyst debate in thread
/scanner [on/off/status] → control scanner delivery
/onchain → on-chain intelligence embed
/gas → color-coded gas price check
/backtest [coin] [strategy] → results as embed + thread
/help → ephemeral command reference

═══════════ RICH EMBED SYSTEM ═══════════

PRICE EMBED:
discord.Embed(title="₿ Bitcoin (BTC)", color=0x10b981 if positive else 0xef4444)
.add_field(name="Price", value="$97,500", inline=True)
.add_field(name="24h", value="+2.3% 📈", inline=True)
.add_field(name="RSI (4h)", value="62.3", inline=True)
.set_thumbnail(url=coin_logo_url)
.set_footer(text="Updated just now • CryptoAgent")

PORTFOLIO: Use code block for alignment:
```

Coin | Value | P&L% ──────────┼──────────────────┼──────────── BTC | $48,750 | +2.6% TOTAL | $58,350 | +1.8%

```
INTERACTIVE BUTTONS (discord.ui.View):
After /price: [📊 Full TA] [🔔 Set Alert] [🔍 Research]
After research: [📔 Journal] [📤 Share in #signals]
```

Alert button → Modal popup for target price input.

REACTION TRACKING:
Scanner posts get reactions: 👀 ✅ 🚫
Track engagement to improve scanner relevance.

========== CHANNEL ROUTING ==========

price_alert → #price-alerts @here
scanner_critical → #market-scanner @here
scanner_medium → #market-scanner (batched every 30 min)
morning_briefing → #morning-briefing @here (pinned)
high_conviction_signal → #signals @everyone
journal_entry → #journal (no ping)

========== CONVERSATIONAL AI ==========

Bot responds when:
- Directly mentioned: @CryptoAgent what's your take on BTC?
- In #crypto-agent: message contains "?" or starts with "analyze/explain/should I
- DM: any message

Public channel: more concise, suggest /commands for detail,
use ephemeral for sensitive data, create threads for long analysis.

Context awareness: include last 5 channel messages in Claude context.

========== BOT PRESENCE ==========

Status rotates every 60 seconds:
"BTC: $97,500 📈" | "Market: Bullish | F&G: 65"
"Scanning 50+ coins..." | "Monitoring [N] alerts"

Status color: Green=online, Yellow=market quiet, Red=critical alert active

========== SCHEDULED POSTS ==========

8:00 AM: Morning briefing → #morning-briefing @here (auto-pin)
Ongoing: Scanner critical = immediate, medium = 30-min batch
7:00 PM: Daily signal digest → #signals (if any high-conviction)
Sunday 6 PM: Weekly performance → #morning-briefing
Midnight UTC: Market close summary → #market-scanner

Deliver: dashboard/app.py, discord_agent/bot.py, commands.py,
embeds.py, alerts.py, scanner.py, permissions.py,
updated main.py, .env.example, README_DISCORD.md

PROMPT: UNIFIED NOTIFICATION ROUTER

```
Create core/notification_router.py

Ensures every alert reaches the right place on every platform.

PRIORITY LEVELS:
1 — CRITICAL: portfolio -10%+, exchange hack, liquidation risk
   → Telegram immediate + Discord DM + #alerts @everyone + Dashboard red banner
   → NO night mode suppression

2 — HIGH: price alert triggered, portfolio -5%+, major news
   → Telegram immediate + Discord @here + Dashboard orange

3 — MEDIUM: scanner finding, opportunity, signal
   → Telegram immediate + Discord no ping + Dashboard badge

4 — LOW: briefings, weekly reviews, informational
   → Telegram scheduled + Discord scheduled + Dashboard feed only

NIGHT MODE (12am-7am):
P1: bypass | P2: queue for 7am | P3+P4: queue for 8am

DEDUPLICATION:
Same event type + symbol within 4 hours = skip.

FORMATTING ADAPTER:
Same data → different format per platform:
format_for_telegram(n) → MarkdownV2 string
format_for_discord(n) → discord.Embed
format_for_dashboard(n) → dict for SSE stream

DATABASE: notification_log (tracks delivery success per platform)
Alert if any platform has >5% failure rate.

/deliverystatus → delivery stats + queue status (on all platforms)
```

PART 7: AIRDROP INTELLIGENCE

Social & On-Chain Reputation Engine

PROMPT 1: CORE AIRDROP ENGINE

```
Create airdrop/ module for tracking and optimizing
on-chain and social reputation for airdrop eligibility.

PHILOSOPHY: This tracks GENUINE protocol usage and helps
build authentic on-chain reputation. Not bot farming.

FILE STRUCTURE:
airdrop/
├── tracker.py       (main + scoring)
├── protocols.py     (protocol registry + criteria)
├── reputation.py    (on-chain reputation)
├── social.py        (social graph analysis)
├── wallet_scorer.py (wallet health)
├── calendar.py      (deadlines + snapshots)
├── task_engine.py   (what to do + when)
└── portfolio.py     (received airdrops tracking)

DATABASE TABLES:
tracked_protocols, airdrop_tasks, wallet_reputation,
social_profiles, airdrop_received, snapshot_tracker

SCORING FORMULA per protocol:
score = (
    transaction_score * 0.20 +
    volume_score * 0.25 +
    consistency_score * 0.25 +   # activity over time
    diversity_score * 0.15 +     # features used
    governance_score * 0.10 +
    social_score * 0.05
)

PRE-LOADED TIER 1 PROTOCOLS:
zkSync Era, Scroll, Linea, Starknet, Monad, Berachain,
MegaETH, Hyperliquid, Kaito AI, EigenLayer

TIER 2: Pendle v2, Morpho expansion, Euler v2, Ekubo, Nostra, Navi, Cetus, Bluefi

TIER 3 (Social): Farcaster apps, Lens v3, Phaver, Interface, Bountycaster

TIER 4 (Infrastructure): Espresso, Lagrange, Brevis, Witness Chain,
Redstone Oracle, Chainlink staking
```

PROMPT 2: WALLET REPUTATION SCORER

```
Create airdrop/wallet_scorer.py + airdrop/reputation.py

WALLET HEALTH SCORE (0-100):

DIMENSION 1: AGE & ACTIVITY (25%)
>2 years: full score | 1-2y: 80% | 6-12m: 50% | <6m: 20%
Active months out of last 12 | Consecutive active streaks
"Consistency > volume. Active 10/12 months beats 500 txns in 1 month."

DIMENSION 2: PROTOCOL DIVERSITY (25%)
Count unique protocols by category:
DEX | Lending | Bridges | Governance | NFT | Staking
Diversity = unique categories × unique protocols

DIMENSION 3: VOLUME & VALUE (20%)
Total lifetime volume + average transaction size.
Projects filter wallets with: high txn count + tiny amounts,
same transaction repeated endlessly, transactions within seconds.

DIMENSION 4: FINANCIAL SOPHISTICATION (15%)
Governance votes, LP provision, advanced DeFi (options/perps/vaults),
staking, contract deployment, limit orders.

DIMENSION 5: SOCIAL PROOF (15%)
ENS name | Gitcoin Passport score | Proof of Humanity
POAP count + diversity | Lens/Farcaster verified address

COMPOSITE OUTPUT:
🔍 WALLET REPUTATION ANALYSIS — 0x1234...5678
OVERALL: 76/100 🟢 STRONG

BREAKDOWN:
Age & Activity:          18/25 ✅ (28 months)
Protocol Diversity:      22/25 ✅ (47 protocols)
Volume & Value:          14/20 ✅ ($340K lifetime)
Financial Sophistication: 12/15 ✅ (governance voter)
Social Proof:            10/15 ⚠️ (no ENS)

GAPS TO ADDRESS:
⚠️ No ENS name (easy win, ~$5/year)
⚠️ Average transaction size $280 (some filters at $500+)

GITCOIN PASSPORT: https://api.scorer.gitcoin.co/registry/score/{address}
Target score >20 for most Gitcoin-using airdrops.
```

```
POAP tracking: https://api.poap.tech/actions/scan/{address}
Governance: snapshot.org GraphQL API for votes by address.

SYBIL RISK FLAGS:
❌ Many transactions within seconds
❌ Identical amounts repeated
❌ Wallet funded directly from exchange (no history before this protocol)
❌ No failed transactions (real users fail)
```

PROMPT 3: SOCIAL REPUTATION ENGINE

```
Create airdrop/social.py

FARCASTER (most important for 2025+):
Free API: https://api.neynar.com (100 req/day free)

FARCASTER SCORE COMPONENTS:
Account Age/FID number (20%): FID <1000 = legendary, <10k = early adopter
Follower Quality (25%): not count but overlap with key accounts
Engagement Rate (20%): replies + recasts + likes per cast vs followers
Network Centrality (20%): connected to protocol founders?
Content Relevance (15%): crypto discussions, governance, technical content

FarcasterAnalyzer:
- get_farcaster_profile(fid_or_username)
- calculate_farcaster_score(fid)
- detect_eligibility_requirements(protocol) → does protocol check Farcaster?
- analyze_social_graph_overlap(my_fid, protocol_fid)
- get_warpcast_frames_activity() → which protocol frames used

LENS PROTOCOL:
Free GraphQL: https://api.lens.xyz
LensAnalyzer: profile handle, follower count, publications, mirrors, collects

KAITO AI:
Increasingly used as airdrop allocation layer.
"Yapper score" = your mindshare contribution.
Track your Kaito score. Protocol mindshare rankings.
Creating analysis content + engaging with protocol discussions builds this.

DISCORD REPUTATION (manual tracking):
TABLE: discord_memberships
Store: server_name, joined_date, current_role, quests_completed
```

```
COMPOSITE SOCIAL SCORE OUTPUT:
Farcaster Score:     72/100 (active, 847 followers)
Lens Score:          45/100 (basic profile)
Kaito Mindshare:     23/100 (some mentions)
Discord Presence:    60/100 (12 servers)
On-Chain Social:     78/100 (ENS, POAPs, attestations)
COMPOSITE:           64/100 🟡 GOOD

KEY SOCIAL GAPS:
⚠️ Kaito score low — create more protocol analysis content
⚠️ Not following tracked protocols on Farcaster
```

PROMPT 4: TASK ENGINE & CALENDAR

```
Create airdrop/task_engine.py + airdrop/calendar.py

CALENDAR: Tracks snapshot dates, TGE events, task deadlines.

SNAPSHOT ALERTS:
30 days before: "⚠️ [Protocol] snapshot in 30 days. Current status: [score]."
7 days before: "🔴 URGENT: snapshot in 7 days. Must complete: [tasks]"
1 day before: "🚨 CRITICAL: snapshot TOMORROW. Final checklist: [items]"

TASK ENGINE:
Analyzes my metrics vs protocol criteria → auto-generates tasks.

TASK EXAMPLES:

For zkSync:
- "Bridge 0.1 ETH to zkSync via official bridge — only 2 bridge txns, need more d
- "Swap on SyncSwap — 8 DEX interactions, target 20+"
- "Deposit to Era Lend — never used lending here"
- "Vote on any zkSync governance — 0 votes currently"

PRIORITIZATION MATRIX:
Priority = (Impact × Urgency) / (Effort × Risk)

DAILY ACTION PLAN:
🪂 DAILY AIRDROP ACTION PLAN

⏰ URGENT (snapshot <14 days):
1. Protocol X — snapshot in 12 days, need 3 more transactions (20 min, $8 gas)
```

```
🔥 HIGH IMPACT TODAY:
2. Starknet — no interaction in 18 days, swap $50 on Ekubo (15 min, $2)
3. Hyperliquid — volume behind target, trade $500 (10 min)

📅 REGULAR MAINTENANCE:
4. EigenLayer — check restaking rewards
5. Kaito — engage with 2 protocol posts

Estimated: 1.5 hours | $24 gas

ANTI-SYBIL ADVICE:
❌ "8 identical $10 swaps today — commonly filtered.
   Better: 2-3 varied transactions, different sizes, different protocols."
❌ "Bridge and immediately withdraw 3x in a row — visible on-chain.
   Better: Leave funds deployed 7+ days."

✅ WHAT LOOKS GENUINE:
Varied transaction sizes | Different protocols each session
Activity spread over weeks | Some failed transactions
Governance participation | Holding assets over time

COMMANDS:
/airdrop tasks | /airdrop complete [task] | /airdrop [protocol]
/airdrop wallet [address] | /airdrop social | /airdrop calendar
```

PROMPT 5: AIRDROP PORTFOLIO TRACKER

```
Create airdrop/portfolio.py

Track received airdrops with full ROI analysis.

RECEIVED AIRDROPS TABLE:
Protocol | Token | Amount | At Receipt | Now | Sold?

ROI CALCULATION:
For each airdrop: track gas spent + time invested.
Calculate effective hourly rate:
"EigenLayer: Received $14,400, 8 hours + $120 gas = $1,785/hour ✅"
"Blast: Received $2,800, 20 hours + $80 gas = $136/hour ⚠️"

DECISION TRACKING:
When you sell: record price and reasoning.
Compare to price 30/90 days later.
Build record of exit timing quality.
```

```
FORWARD ESTIMATES:
Protocol    | Likelihood | Est. Min | Est. Max | Priority
zkSync      | 85%        | $500     | $8,000   | 5/5
Scroll      | 75%        | $300     | $4,000   | 4/5
Monad       | 70%        | $500     | $10,000  | 5/5

Conservative total: $4,900 | Optimistic: $62,000 | Most likely: $18,000

/airdrop portfolio → full received + estimated forward value
```

PROMPT 6: AIRDROP DASHBOARD INTEGRATION

```
Create airdrop/tracker.py (main module) and integrate everything.

TELEGRAM MASTER COMMAND: /airdrop
Shows:
🪂 AIRDROP INTELLIGENCE HUB

WALLET REPUTATION: 76/100 🟢
SOCIAL SCORE: 64/100 🟡

TRACKED PROTOCOLS: 18
├ ✅ Criteria met: 8
├ 🟡 Close (1-2 tasks): 6
└ ❌ Need work: 4

SNAPSHOT ALERTS:
🔴 Protocol X — 12 days
🟡 Protocol Y — 28 days

TODAY'S TOP PRIORITY:
Bridge 0.1 ETH to zkSync, swap on SyncSwap.
20 min | $8 gas

AIRDROP PORTFOLIO: $38,400 current value

AUTOMATED:
Daily 8am: today's task list + snapshot alerts
Weekly Sunday: wallet score + social score update + strategy
Real-time: new airdrop announced → immediate alert
New protocol detected → "Add to tracking?" prompt

NEW PROTOCOL DETECTION:
```

```
Monitor DeFiLlama new protocols with no token + notable backing.
Alert: "🆕 NEW OPPORTUNITY: Protocol X, Backed by [VCs],
TVL growing $Xm in X days, ~2 hours setup, $30 gas to qualify.
[Add] [Research First] [No]"

DASHBOARD /airdrops page:
Tabs: [Overview] [Protocols] [Tasks] [Calendar] [Reputation] [Social] [Portfolio]
Tasks tab: Kanban board — [To Do Today] [This Week] [Someday] [Done]

CLAUDE CONTEXT INJECTION:
"AIRDROP STATUS: Wallet 76/100, Social 64/100, 18 protocols active,
Snapshot imminent: Protocol X (12 days), Today's priority: zkSync tasks"
→ Claude proactively mentions: "You haven't touched Starknet in 18 days."
```

PART 8: THE LIVING AGENT

Orchestration, Skills, Memory & Evolution

PROMPT 1: THE COGNITIVE ARCHITECTURE

```
Create core/cognitive_loop.py

Build a cognitive loop that mirrors how expert human analysts think.

THE HUMAN EXPERT MODEL (10-phase loop, runs every 15 minutes):

PHASE 1: PERCEPTION
Only process CHANGES since last cycle (not full world state):
perception = {
  'price_changes': get_significant_changes(),
  'news_events': get_new_news(),
  'onchain_events': get_new_onchain(),
  'social_shifts': get_social_changes(),
  'portfolio_changes': get_portfolio_delta()
}

PHASE 2: SIGNIFICANCE FILTERING
significance_score = (magnitude_of_change * 0.35 +
                      relevance_to_portfolio * 0.30 +
                      historical_precedent * 0.20 +
                      recency_weight * 0.15)
```

```
Only events above threshold enter working memory.

PHASE 3: BELIEF UPDATING
Agent maintains probabilistic belief state:
beliefs = {
  'btc_trend': {'direction': 'up', 'confidence': 0.72},
  'market_regime': {'type': 'bull', 'confidence': 0.68},
  'eth_relative_strength': {'vs_btc': 'weak', 'confidence': 0.61}
}
Bayesian updating: new_confidence = (prior × likelihood) / normalizer
Confidence <0.4: UNCERTAIN — flag for investigation
Confidence <0.3: INVALIDATED — alert user, update strategy

PHASE 4: UNCERTAINTY MAPPING
uncertainty_map = {
  'questions': [
    {'question': 'Why is ETH underperforming?', 'priority': 0.8},
    {'question': 'Is BTC breakout real?', 'priority': 0.9}
  ]
}
These questions drive self-directed research.

PHASE 5: ATTENTION ALLOCATION
Budget: 10 units per cycle.
Active positions: 3 | Unanswered questions: 3
Best opportunity: 2 | Risk monitoring: 1 | Learning: 1

PHASE 6: AUTONOMOUS REASONING
Agent CHOOSES which analysis to run based on what it's investigating.
"Why is ETH underperforming?" → runs ETH/BTC ratio, on-chain, news, derivatives.

PHASE 7: HYPOTHESIS FORMATION
hypothesis = {
  'statement': 'ETH underperformance driven by L2 rotation, not fundamental weakn
  'confidence': 0.65,
  'testable_prediction': 'ETH/BTC ratio stabilizes within 5 days',
  'expires_at': timestamp + 7_days,
  'outcome': None  # filled in later
}

PHASE 8: ACTION DECISION
Only notify if: belief confidence flips, hypothesis confirmed/denied,
risk threshold breached, high-conviction opportunity, or user has pending questio
MOST CYCLES: No notification. Work silently.

PHASE 9: CYCLE REFLECTION
Log: belief state changes, hypotheses formed/tested, questions answered.
```

```
Raw material for evolution engine.

DATABASE TABLES:
belief_state | hypotheses | uncertainty_map | cognitive_cycles | attention_histor
```

PROMPT 2: THE SKILL SYSTEM

```
Create intelligence/skill_system.py

A skill is a PATTERN OF REASONING proven effective in a specific context,
stored for automatic recall when that context recurs.

SKILL ANATOMY:
skill = {
  'name': 'ETH RSI Bounce Identification',
  'category': 'technical_analysis',
  'trigger_conditions': {
    'coin': ['ETH'], 'rsi_below': 38,
    'volume_trend': 'declining', 'timeframe': '4h',
    'market_regime': ['bull', 'neutral']
  },
  'reasoning_template': """
    1. Check if price above 200 SMA (must be)
    2. Look for RSI divergence (higher RSI lows while price makes lower lows)
    3. Check funding rate (should be neutral/negative)
    4. Check ETH/BTC ratio trend
    5. If 3/4 met: high conviction bounce
    Entry: current price ±1.5% | Stop: below the divergence low
    Target: previous resistance or 0.618 fib
  """,
  'times_applied': 23, 'win_rate': 0.696,
  'expected_value': 0.066, 'version': 3
}

PRE-LOADED SKILL LIBRARY (28 skills):
Technical: RSI Bounce, Volume Breakout, BB Squeeze, MACD Divergence,
           Support/Resistance Flip, Higher TF Alignment, Funding Extremes,
           Options Max Pain, Pre-expiry Pattern, Post-news Reaction

Macro: Risk-off Cascade, DXY Inflection, Fed Decision Response,
       Correlation Break, Sector Rotation Early Detection

On-chain: Exchange Inflow Pressure, Whale Accumulation,
          Smart Money Divergence, Mempool Congestion, Protocol TVL Flow
```

```
Airdrop: Snapshot Preparation, Sybil Avoidance, Governance Timing, Batching

Psychology: Post-loss Cooling, FOMO Detection, Entry Sizing, Profit-taking

SKILL ACQUISITION (4 methods):
1. PATTERN DISCOVERY: Agent finds pattern in historical data, proposes skill
2. MANUAL CREATION: /skill create "when BTC drops 5% on low volume near 50 SMA...
3. OUTCOME ABSTRACTION: After correct analysis, offer to save as reusable skill
4. SKILL IMPORT: Pre-built packs (Day Trading / Swing Trading / DeFi / Airdrop)

SKILL EVOLUTION:
When win rate drops >10% over last 10 applications:
Step 1: Autopsy — analyze failed cases for common factor
Step 2: Propose refinement with backtested improvement
Step 3: A/B test old vs new version (10 applications each)
Step 4: Keep better performer, archive old version

SKILL RETIREMENT:
If consistently <50% win rate over 15 applications:
[Refine] [Retire] [Pause 30 days] [Investigate failures]

COMMANDS:
/skills → library with performance stats
/skills technical → filtered by category
/skills top → 5 highest-performing
/skill apply ETH → find and apply all applicable to ETH
/skill history [name] → full performance history
```

PROMPT 3: THE MEMORY ARCHITECTURE

```
Create intelligence/memory_architecture.py

7-layer memory system mirroring human cognition:

LAYER 1: SENSORY BUFFER (60 seconds)
Raw market data stream. 95% discarded.
Significance filter passes to Layer 2:
Price >2% change | Volume >3x | News impact >6 | On-chain >$10M

Filter is LEARNED: adjusts based on what user actually acted on.

LAYER 2: WORKING MEMORY (current session)
Capacity: 7 items (mirrors human limit).
```

When full: least relevant item consolidates to Layer 3.
Always-available, zero-latency context for active conversation.

LAYER 3: SHORT-TERM (7-30 days, unlimited capacity)
TABLE: short_term_memory (id, content, importance, access_count, decay_rate)
Nightly consolidation at 3am:
Accessed >3 times → long-term | Confirmed pattern → long-term
0 accesses + old → DISCARD | Uncertain → keep, reduce importance

LAYER 4: LONG-TERM (permanent, compressed)
Not events — PATTERNS:
Not "Feb 14 ETH fell 8%" but "ETH tends to fall Valentine's Day weekend (3/4 year

Types:
EPISODIC: "March 2024 ETH sell-off during BTC ETF launch — recovered in 11 days"
SEMANTIC: "Hyperliquid has <1ms settlement, fastest perp DEX available"
PATTERN: "BTC dumps 3-5% week before quarterly options expiry (4/5 recent quarter
RELATIONSHIP: "DXY strengthening → BTC weakness → ETH extra weakness → altcoin ca

LAYER 5: PROCEDURAL (skill system — automatic, pattern-triggered)
LAYER 6: SEMANTIC (knowledge base — protocol facts, market mechanics)
LAYER 7: EPISODIC (personal history — every analysis, prediction, conversation)

KEY OPERATIONS:

recall(query, context, limit):
1. Check working memory (instant)
2. Search short-term (fast)
3. Search long-term patterns (medium)
4. Search episodic history (comprehensive)
Ranked by: (relevance × recency × importance × confidence)

associate(memory_1, memory_2, relationship_type):
Builds associative network. ETH recall → triggers memory of last ETH trade
→ triggers that market regime → triggers lesson learned.

consolidate() (nightly):
Review 24h → find patterns → compress to long-term → discard low-value.

forget(memory_id, reason): Intentional obsolescence.
"2021 bull market patterns removed — current structure differs significantly."

COMMANDS:
/memory → full memory status (all 7 layers)
/memory search "ETH bull flag" → multi-layer search with ranked results
/memory forget "2021 bull patterns" → archive obsolete patterns
/memory important → highest-weight memories

PROMPT 4: THE EVOLUTION ENGINE

```
Create intelligence/evolution_engine.py

DISTINCTION: Accumulating data ≠ learning.
A database collects for years and doesn't change.
An expert analyst after years makes fundamentally better decisions.
We are building the latter.

DIMENSION 1: PREDICTION CALIBRATION
Track every prediction's stated confidence vs actual outcome.
When agent says 70% confident: should be right 70% of the time.

Miscalibration patterns:
OVERCONFIDENCE: Says 80%, right 55% → reduce confidence by miscalibration factor
UNDERCONFIDENCE: Says 50%, right 73% → increase confidence scores
DOMAIN-SPECIFIC: Well-calibrated on BTC, overconfident on altcoins → domain corre
TIME HORIZON: Accurate at 24h, miscalibrated at 7d → horizon-specific corrections

Monthly calibration curve. Automatic correction factor applied every Sunday.

DIMENSION 2: REASONING QUALITY SCORING
Correct predictions from bad reasoning = lucky. Wrong from good reasoning = bad l
What matters long-term: REASONING QUALITY.

Evaluate each prediction post-resolution (1-5 each):
1. Were key factors identified correctly?
2. Was uncertainty appropriately expressed?
3. Was the counter-thesis considered?
4. Was evidence properly weighted?
5. Was conclusion proportional to evidence?

Track quality separately from outcome. Improve lowest-scoring dimensions.

DIMENSION 3: MENTAL MODEL UPDATING
Maintain explicit mental models with confidence levels:
'btc_as_risk_asset': {confidence: 0.71, supporting/contradicting evidence}
'eth_gas_demand_model': {confidence: 0.58, note: "weakening post-EIP4844"}
'altcoin_beta_model': {confidence: 0.76, exceptions: ["during BTC dominance drops

When contradicted 3+ times: propose update with evidence summary.

DIMENSION 4: SELF-DIRECTED LEARNING
```

```
TABLE: learning_queue (topic, why_needed, priority, status, knowledge_gained)

Triggers for adding to queue:
- Prediction failure with unknown cause
- User asks question agent can't answer
- New market structure detected
- Skill consistently underperforms

Learning process: Research → Synthesize → Hypothesize → Integrate → Notify user

DIMENSION 5: PERSONALITY EVOLUTION
Track and adapt:
Response length (what gets engagement vs dismissal?)
Proactivity level (too many alerts? too few?)
Confidence expression (ranges vs point estimates?)
Time preferences (when is user most responsive?)
Depth preferences (which topics want technical depth?)

All adapts automatically from behavior observation.

DIMENSION 6: COLLECTIVE INTELLIGENCE
Monthly conversation pattern analysis:
What's asked repeatedly → include proactively in briefings.
What's dismissed consistently → reduce or eliminate.
What triggers longest engagement → provide more.
What leads to journal entries → these insights matter most.

WEEKLY EVOLUTION REPORT:
Predictions: X correct/total (calibration good/off)
Skills applied: X (best/worst performer)
Memory: X new patterns, X consolidated, X discarded
Learning: X completed, X started
Mental model updates: [list]
Adaptation this week: [behavioral changes made]
My assessment: [agent's honest self-evaluation paragraph]
```

PROMPT 5: THE MASTER ORCHESTRATOR

```
Create core/master_orchestrator.py

The conductor that coordinates all agents, skills, memory, workflows
as a unified intelligence. Best orchestration is invisible.

ROUTING INTELLIGENCE (8 steps for every input):
```

```
Step 1: INTENT CLASSIFICATION
question_factual | question_analytical | question_strategic
action_request | emotional_input | market_event
scheduled_task | learning_trigger | skill_trigger

Step 2: CONTEXT ASSEMBLY
For question_analytical: relevant memories + skills + market data
+ portfolio state + predictions about this topic + belief state

Step 3: AGENT SELECTION

SPECIALIST AGENTS:
TechnicalAnalyst: charts, patterns, indicators, levels
FundamentalAnalyst: protocol analysis, tokenomics, competitive position
MacroStrategist: macro environment, institutional flows, regulatory
RiskManager: position sizing, portfolio heat, drawdown, correlation
AirdropStrategist: protocol eligibility, tasks, wallet reputation
PsychologyCoach: emotional state, behavioral patterns, bias detection
ResearchAgent: deep dives, due diligence, competitor comparison
ThreeAnalystDebate: contested high-stakes decisions

Step 4: SKILL ACTIVATION
Scan all skill triggers against context. Apply matching skills.

Step 5: MEMORY RETRIEVAL
Query all 7 layers for relevant past experience.

Step 6: RESPONSE SYNTHESIS
Chosen agent + memory + skill outputs + calibration factors.

Step 7: QUALITY CHECK
Calibrated correctly? Right length for this user? Conflicts with recent statement

Step 8: DELIVERY + LOGGING
Log: intent, agents, skills, memories used. Track outcome.

COORDINATION PATTERNS:
SEQUENTIAL: TechnicalAnalyst → RiskManager → ThreeDebate
PARALLEL: MacroStrategist ‖ FundamentalAnalyst ‖ TechnicalAnalyst
ITERATIVE: ResearchAgent → Technical → ResearchAgent (deeper)
ADVERSARIAL: Bull agent vs Bear agent → RiskManager decides

WORKFLOW ENGINE (integrated):
workflow = {
  'name': 'Morning Market Preparation',
  'trigger': {'type': 'schedule', 'time': '07:45'},
```

```
  'steps': [
    {'id': 'warm_cache', 'agent': 'data_fetcher', 'timeout': 30, 'on_failure': 'u
    {'id': 'scan_overnight', 'depends_on': ['warm_cache']},
    {'id': 'run_cognitive_cycle', 'depends_on': ['warm_cache']},
    {'id': 'activate_skills', 'depends_on': ['warm_cache']},
    {'id': 'generate_briefing', 'depends_on': ['scan_overnight', 'run_cognitive_c
    {'id': 'synthesize_with_claude', 'depends_on': ['generate_briefing'],
     'context_includes': ['market_data', 'overnight_events', 'active_beliefs',
                          'skill_signals', 'portfolio_state', 'memory_relevant']}
    {'id': 'deliver', 'depends_on': ['synthesize_with_claude']}
  ]
}

NATURAL LANGUAGE WORKFLOW CREATION:
/workflow create "Every time BTC drops 5% in a day, analyze if it's a buying
opportunity and send me a full report with position sizing"
→ Agent parses, creates formal workflow, confirms with user.

/thinking → shows orchestrator's reasoning for last response (full transparency)
/orchestrator → status dashboard
```

PROMPT 6: INTEGRATION — THE LIVING SYSTEM

```
Integrate all new systems into existing codebase as a coherent whole.

THE INFORMATION FLOW:

WORLD EVENTS
    ↓
SENSORY BUFFER (filters 95% of noise)
    ↓
COGNITIVE LOOP (every 15 min)
    ↓
BELIEF STATE UPDATE
    ↓
SKILL TRIGGER CHECK ← SKILL LIBRARY
    ↓
MEMORY RETRIEVAL ← 7-LAYER MEMORY ARCHITECTURE
    ↓
MASTER ORCHESTRATOR
    ↓              ↓
SPECIALIST      WORKFLOW
AGENTS          ENGINE
    ↓              ↓
```

```
RESPONSE        SCHEDULED
SYNTHESIS       ACTIONS
    ↓               ↓
NOTIFICATION ROUTER
    ↓        ↓        ↓
TELEGRAM  DISCORD  DASHBOARD
    ↓
OUTCOME TRACKING
    ↓
EVOLUTION ENGINE
    ↓
SKILL REFINEMENT + CALIBRATION UPDATE + MEMORY CONSOLIDATION
    ↓
BACK TO WORLD EVENTS (smarter than before)
```

INITIALIZATION ORDER IN main.py:

1. Database (all tables)
2. Memory Architecture
3. Skill System (loads from DB + memory)
4. Belief State (loads from DB)
5. Evolution Engine (registers with skill system)
6. Master Orchestrator (connects all agents)
7. Cognitive Loop (registers with orchestrator)
8. Workflow Engine (loads saved workflows)
9. Notification Router
10. Data Services (price, news, onchain, social)
11. Telegram Bot
12. Discord Bot
13. Dashboard
14. Start cognitive loop (background)
15. Start scheduled workflows
16. Log: "Agent initialized. Skills: X. Memory: X patterns. Beliefs: X. Ready."

UPGRADED CONTEXT INJECTION (every Claude call):

STATIC: Agent identity, trading rules, skill library summary, mental models
DYNAMIC: Current belief state, working memory, retrieved relevant memories,
active triggered skills, applicable hypotheses, portfolio state,
calibration factors, personality preferences
EPISODIC: Similar past situations, relevant prediction history, recent lessons

COGNITIVE HEALTH DASHBOARD (new dashboard section):
Memory: Working 5/7 slots | Short-term: 4,247 | Long-term: 312 patterns
Beliefs: High confidence: 14 ✅ | Uncertain: 6 ⚠️ | Needs review: 3 ❌
Skills: Healthy: 21 ✅ | Under review: 5 ⚠️ | Flagged: 2 ❌
Cognitive Loop: Last cycle 8 min ago | 94 cycles today
Evolution: Calibration error 4.2% | Skill improvement +3.1% this month

```
DELIVER:
1. core/cognitive_loop.py
2. intelligence/skill_system.py
3. intelligence/memory_architecture.py
4. intelligence/evolution_engine.py
5. core/master_orchestrator.py
6. Updated main.py (correct initialization order)
7. Updated storage/database.py (all new tables)
8. Updated core/context_builder.py (inject memory + beliefs + skills)
9. Updated bot/handlers.py (route through orchestrator)
10. dashboard/cognitive_health section
11. Complete requirements.txt additions

Here is my current codebase: [PASTE ALL YOUR FILES]
```

🔑 MASTER INTEGRATION PROMPT

Use This After Each Major Section

```
I have now built [SECTION NAME] for my crypto intelligence agent.
Before I continue to the next level, I need you to:

1. CHECK ALL IMPORTS
   Make sure every file imports correctly from every other file.
   List any missing imports.

2. CHECK DATABASE CONSISTENCY
   Review all table definitions.
   Ensure no duplicates, consistent foreign keys,
   all functions referenced actually exist.

3. CHECK THE SCHEDULER
   List every scheduled task:
   - Trigger (time/event)
   - Function called
   - Approximate run time
   - Conflicts with other tasks

4. CHECK ERROR HANDLING
   Top 5 most likely failure points.
   Ensure each has proper handling.
```

```
5. STARTUP SEQUENCE
   Correct initialization order for main.py.

6. PERFORMANCE AUDIT
   Functions making API calls without caching.
   Functions called too frequently.
   Suggest optimizations.

7. COST ESTIMATE
   Monthly Anthropic API cost at typical usage.
   How to stay under $50/month.

8. FINAL REQUIREMENTS.TXT
   Every library needed with version numbers.

Here is my current codebase: [PASTE ALL FILES]
```

📦 ENVIRONMENT VARIABLES REFERENCE

```
# Core
TELEGRAM_BOT_TOKEN=from_botfather
ANTHROPIC_API_KEY=sk-ant-...
ALLOWED_USER_ID=your_telegram_id

# Market Data
BINANCE_API_KEY=optional_for_advanced
ETHERSCAN_API_KEY=free_at_etherscan.io
COINGECKO_API_KEY=optional_free_tier

# News
CRYPTOPANIC_API_TOKEN=free_registration

# Dashboard
DASHBOARD_PASSWORD=choose_strong_password
DASHBOARD_PORT=8080
DASHBOARD_SECRET_KEY=random_32_chars

# Discord
DISCORD_TOKEN=from_dev_portal
DISCORD_GUILD_ID=your_server_id
DISCORD_GENERAL_CHANNEL=channel_id
DISCORD_ALERTS_CHANNEL=channel_id
DISCORD_SCANNER_CHANNEL=channel_id
DISCORD_BRIEFING_CHANNEL=channel_id
```

```
DISCORD_JOURNAL_CHANNEL=channel_id
DISCORD_SIGNALS_CHANNEL=channel_id
DISCORD_ADMIN_USER_ID=your_discord_id

# Airdrop
GITCOIN_SCORER_API_KEY=free_at_scorer.gitcoin.co
NEYNAR_API_KEY=free_tier_farcaster
POAP_API_KEY=free_at_poap.tech

# Advanced
GROQ_API_KEY=free_tier_whisper_transcription
LUNARCRUSH_API_KEY=free_tier

# Security
ENCRYPTION_KEY=auto_generated_on_first_run
```

🗺️ COMPLETE SYSTEM MAP

```
FOUNDATION (1-10): Telegram bot, Portfolio, Alerts, Market data,
                   Journal, Briefings, Advanced alerts, Wallet watcher,
                   AI advisor, Security

INTELLIGENCE (11-20): Scanner, On-chain, News+Sentiment, Research+Optimizer,
                      TA (9 indicators), Performance tracking, Interactive UI,
                      Web dashboard, Workflows, Master orchestrator

PROFESSIONAL (21-30): Quant models+Backtesting, DeFi+Gas, ML signals,
                      Social intelligence, Institutional flows, Multi-chain,
                      Voice interface, Kelly+Position sizing, Options intelligenc
                      Three-analyst debate

ELITE (31-40): Event predictor, Macro correlations, Task queue, Security hardenin
               Performance attribution, Trade proposals, REST API+TradingView,
               Simulation environment, Crypto academy, Unified intelligence hub

FRONTIER (5-10): Alternative data, Arbitrage scanner, Behavioral psychology,
                 Self-improving prompts, Infrastructure decisions, The final form

INTERFACES: Web Dashboard (10 pages, SSE real-time) + Discord (native, slash comm

AIRDROP: Protocol registry, Wallet scorer, Social reputation, Task engine,
         Calendar, Portfolio tracker, Dashboard page, Automated daily tasks

LIVING AGENT: Cognitive loop, Skill system (28+ skills), 7-layer memory,
```

Total build time following all prompts at 2-3 hours/day: approximately 3-6 months for complete system. Start with Levels 1-3. Make them rock solid. Build from there. The system that compounds in value every day you use it.